algorithms

MDPI

Article

Cardiovascular Disease Risk Prediction Utilizing Two-Tier Classification Framework Optimized with Adapted Variable Neighborhood Search Algorithm

Saramma John Villoth [1], Petar Dabic [2], Tamara Zivkovic [3], Miodrag Zivkovic [3], Svetlana Andjelic [3], Milos Mravik [3], Vladimir Simic [4,5,6], Mahmoud Abdel-Salam [7] and Nebojsa Bacanin [3,8,*]

1 University of Technology and Applied Sciences, Nizwa 611, Oman; sara.john@utas.edu.om
2 Institute for Cardiovascular Diseases Dedinje, Heroja Milana Tepica 1, 11000 Belgrade, Serbia
3 Faculty of Informatics and Computing, Singidunum University, Danijelova 32, 11000 Belgrade, Serbia; tzivkovic@singidunum.ac.rs (T.Z.); mzivkovic@singidunum.ac.rs (M.Z.); mmravik@singidunum.ac.rs (M.M.)
4 Faculty of Transport and Traffic Engineering, University of Belgrade, Vojvode Stepe 305, 11010 Belgrade, Serbia
5 Faculty of Engineering, Dogus University, Umraniye, 34775 Istanbul, Türkiye
6 Department of Computer Science and Engineering, College of Informatics, Korea University, Seoul 02841, Republic of Korea
7 Faculty of Computers and Information Science, Mansoura University, Mansoura 35516, Egypt; mahmoud20@mans.edu.eg
8 Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences (SIMATS), Thandalam, Chennai 602105, Tamil Nadu, India
* Correspondence: nbacanin@singidunum.ac.rs

Abstract

Accurately assessing a patient's likelihood of developing cardiovascular conditions is essential for proper case classification and for ensuring timely, targeted medical intervention. To address this need, the present study employs a carefully optimized machine learning framework to predict such risks within cardiology settings. A hybrid architecture is proposed that combines convolutional neural networks (CNNs) with cutting-edge gradient boosting classifiers, namely CatBoost and LightGBM, whose performance is further enhanced by metaheuristic optimization. The system adopts a two-layer design capable of capturing complex data structures while supporting accurate classification of cardiac patients and their risk of developing cardiovascular disease. Extensive evaluation on real-world data confirms the framework's effectiveness for binary classification, with the best models reaching an accuracy of slightly over 92%. To complement predictive performance, explainable AI methods were applied to clarify model decisions, yielding practical insights that can guide future data collection strategies and improve diagnostic precision.

Keywords: cardiovascular diseases; risk prediction; classification; CNN; CatBoost; LightGBM; swarm intelligence; variable neighborhood search; metaheuristics optimizers

Check for updates

Academic Editor: José Manuel García-Nieto

Received: 19 December 2025
Revised: 20 January 2026
Accepted: 30 January 2026
Published: 5 February 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Cardiovascular diseases (CVDs) are the leading cause of morbidity and mortality worldwide, accounting for an estimated 19.8 million deaths in 2022, or roughly one-third of all global deaths [1]. Despite advances in prevention and treatment, the global burden of CVD continues to rise, largely driven by population aging, increasing prevalence of diabetes and obesity, and persistent tobacco use. Beyond mortality, CVD is a major contributor

to disability-adjusted life years (DALYs), highlighting the urgent need for accurate risk stratification and early preventive strategies [2].

To address this challenge, several multivariable risk scores have been developed over the past four decades. The Framingham Risk Score, derived from the landmark Framingham Heart Study, was one of the earliest models to integrate age, sex, cholesterol, blood pressure, smoking, and diabetes into a standardized 10-year risk estimate for coronary heart disease [3–5]. Since then, regional adaptations have emerged, such as the SCORE system in Europe, which predicts cardiovascular mortality, and the ASCVD Pooled Cohort Equations in the United States, which estimate 10-year risk of both fatal and non-fatal cardiovascular events [6–8]. More recently, QRISK3 models have been developed in the United Kingdom to incorporate additional variables such as ethnicity, socioeconomic status, and comorbidities, reflecting the need for greater inclusivity and precision in risk prediction [9].

These risk scores have been widely implemented in clinical practice and guidelines, providing a framework for preventive interventions such as statin initiation, antihypertensive therapy, and lifestyle modification. However, their predictive performance is not without limitations. Numerous validation studies demonstrate reasonable discrimination (c-statistics in the range of 0.70–0.80) but variable calibration across populations, often overestimating risk in low-incidence settings and underestimating risk in high-risk groups such as younger individuals, women, and ethnic minorities [9,10]. Such discrepancies underscore the limitations of relying solely on traditional risk factors and point to the need for more flexible, individualized approaches.

In this context, artificial intelligence (AI) and machine learning (ML) have emerged as promising tools to enhance cardiovascular risk prediction [11,12]. By leveraging large-scale electronic health records, imaging data, and novel biomarkers, AI models can capture complex, nonlinear relationships that conventional risk scores may overlook [13–18]. Recent studies suggest that ML-based models can improve discrimination and calibration compared to traditional scores, while also enabling dynamic, patient-specific risk estimation [19].

A key limitation of AI lies in the many challenges it needs to address. These encompass issues ranging from poor data quality and unsuitable models to ineffective hyperparameter choices. For example, utilizing biased or low-quality data inevitably produces unreliable results, underscoring the importance of high-quality datasets for robust AI models. Likewise, selecting the right ML algorithm is equally important, since performance can vary widely depending on the dataset and the problem at hand. Although model choice plays an important role, it is equally critical to define a clear research question and select a modeling approach that is well suited to the objectives and characteristics of the problem under investigation. Model's hyperparameters also play a decisive role in model effectiveness and therefore require careful optimization. According to Wolpert's no free lunch (NFL) theorem [20], no single model can deliver optimal results across all classification problems, meaning solutions must be tailored to specific tasks. Nevertheless, optimizing hyperparameters is notoriously difficult and is widely acknowledged as an NP-hard problem. As the number of parameters and data dimensions increases, the computational cost of finding the best configuration can become prohibitive. Conventional optimization approaches often fail to produce timely or reliable outcomes under such complexity, constraining their applicability in large-scale or intricate scenarios.

One effective way to handle this challenge is through metaheuristic optimization methods, which are able to navigate vast search spaces to deliver near-optimal solutions. Rather than aiming for exact answers, which are often infeasible in complex, real-world contexts, these techniques focus on producing strong approximations with practical effi-

ciency. In the realm of hyperparameter tuning, metaheuristics mitigate the limitations of traditional optimization by offering adaptable and scalable strategies. As a result, they play a crucial role in enhancing the performance of AI systems across a wide range of applications. Bayesian optimization methods, such as those implemented in Optuna, represent a popular alternative for hyperparameter tuning and have shown strong performance in many machine learning applications. However, their inclusion as an additional benchmark is left for future work due to the already extensive experimental scope of this study.

This study introduces a two-tier approach for advancing cardiovascular disease prediction, building on a two-stage framework inspired by prior research [21,22]. In this framework, the initial phase leverages a deep learning (DL) convolutional neural network (CNN) to extract high-level features from patient data. Extending insights from earlier works [21,22], which demonstrated notable gains by replacing the CNN's final dense layer with ensemble classifiers, the proposed system routes the extracted features into a second stage that integrates both CatBoost [23] and LightGBM [24] classifiers. To maximize predictive performance, metaheuristic algorithms are applied across both stages to fine-tune hyperparameters, ensuring optimal synergy between deep feature extraction and ensemble classification. This integration capitalizes on the representational power of CNNs and the robustness of boosting techniques, while metaheuristics ensure near-optimal parameter choices. Ultimately, the layered design produces a system that combines the adaptability of DL with the reliability of ensemble methods, yielding enhanced accuracy and efficiency in predicting cardiovascular diseases. The introduction of such a novel two-tier framework, proposed here for the first time in the literature, represents an original contribution to the field of cardiovascular disease prediction.

In this study, an adapted variant of the variable neighborhood search (VNS) [25] is employed to fine-tune the hyperparameters of the models within the proposed framework, with the objective of maximizing predictive accuracy. The choice of VNS was informed by extensive comparisons with alternative optimization strategies and is consistent with the NFL theorem [20], which asserts that no single optimization approach can universally dominate across all problem domains. Although more recent optimization methods have been introduced, preliminary experiments revealed that even the baseline form of VNS delivered highly competitive results on smaller-scale evaluations. This strong empirical performance established VNS as a suitable foundation for further refinement, allowing it to be adapted for greater robustness and efficiency. The enhanced VNS thus plays a central role in boosting the reliability and effectiveness of the framework for cardiovascular disease prediction. Lightweight tuning strategies were applied to CNN, LightGBM, and CatBoost models. This design choice ensures that the resulting models can be efficiently deployed on resource-constrained platforms, like TinyML environments and low-power devices including Arduino and ESP32 chips.

Building on the preceding discussion, the primary contribution of this research is a two-tier framework for CVD risk prediction that:

- Separates deep feature extraction from classifier optimization to improve search efficiency and stability;
- Employs an adapted VNS-based optimizer specifically configured for this staged architecture;
- Demonstrates the effectiveness of this design on real-world cardiovascular data.

The novelty of the proposed framework does not lie in introducing a fundamentally new ML paradigm, but rather in the integration and adaptation of established techniques into a unified, two-tier optimization framework tailored specifically for CVD risk prediction. Although CNNs, gradient-boosting classifiers, and metaheuristic optimization methods are well known individually in the ML literature, their staged combination, optimization strat-

egy, and systematic application to CVD risk modeling constitute the primary contribution of this work.

The structure of this paper is as follows: Section 2 reviews the theoretical foundations and background concepts of the applied methods. Section 3 details the core algorithm employed in the experimental framework. The configuration of the simulation environment is described in Section 4, providing sufficient detail to enable reproducibility. Section 5 reports the outcomes of the experiments and offers a thorough discussion of their implications, while Section 6 provides statistical analysis and explainability of the outcomes. Lastly, Section 7 highlights the main contributions and insights, and outlines potential directions for future investigations.

2. Related Works

AI has already demonstrated clinical utility across diverse medical domains, including transplantation care and complication management, where integrated and interpretable AI systems are increasingly emphasized [26]. CVD prediction has increasingly benefited from ML and AI, especially those capable of handling complex, high-dimensional datasets. One systematic review of AI-driven CVD risk prediction models highlighted significant methodological heterogeneity among studies, showing that among 212 articles, only 36% underwent external validation, and just 19% were independently validated [27]. This underscores a gap between model development and broader clinical applicability, despite ML's demonstrated capability to reduce classification errors and enhance diagnostic reliability.

AI models specifically designed to handle time-to-event outcomes are relatively underutilized. A 2024 systematic review by Teshale et al. found that although ML and deep learning methods are increasingly used for survival analysis, their handling of right-censored data remains limited. Few models include social determinants of health or gender stratification, and many lack comprehensive explainability frameworks [28]. This sentiment is echoed in broader reviews, which consistently call for improved model validation, transparency, and adaptability across diverse population contexts [29].

Despite these challenges, evidence supports the superior predictive accuracy of ML models over traditional risk scores. A meta-analysis of ML versus conventional models using electronic health records found pooled AUCs of 0.865 for random forests and 0.847 for deep learning, compared to 0.765 for traditional risk scores like QRISK3 or ASCVD [30]. In a study of over 350,000 patients, Weng et al. showed that gradient boosting, random forests, and neural networks trained on routine clinical variables outperformed Framingham and QRISK in both reclassification and discrimination. Furthermore, they identified additional high-risk patients who could benefit from preventive treatments [31]. Large-scale work using automated ML on UK Biobank data showed good gains and emphasized that the incremental benefit often arises from richer feature sets rather than model complexity alone [32]. Similarly, deep learning models such as DeepSurv have shown improvements in metrics like C-index and Brier Score over conventional Cox models, even translating to clinical efficiency gains by lowering the number needed to treat [33].

Recent paradigm-shifting reviews highlight that ML and AI enable a multifaceted, precision medicine approach to CVD risk assessment. ML algorithms can model nonlinear relationships among diverse variables, and offer individualized risk insights that traditional models miss [34,35]. Yet, the authors emphasize enduring challenges: biased datasets, limited external validation, inconsistent reporting standards, and barriers to clinical integration must be addressed before these tools can fundamentally reshape CVD prevention.

Lifestyle and health-behavior data recorded in routine care, such as smoking status, alcohol use, physical activity or diet, have been incorporated into ML risk models and shown to enhance prediction, particularly for reclassifying individuals at intermediate risk. Recent

studies and reviews indicate that when richer behavioral data are available, ML models can better stratify risk and help target preventive services. Nonetheless, the heterogeneity and missingness of behavior and preventive-service coding across systems limit generalizability unless explicit missing-data strategies and bias assessments are applied [36–39].

This limitation highlights the pressing need for rigorously validated, AI-driven frameworks, ideally subject to independent evaluation, that can reliably forecast cardiovascular risk. Such frameworks should not merely chase marginal accuracy improvements, but instead aim for strong generalizability, greater transparency in decision-making, and practical relevance in clinical settings. Approaches of this kind hold the potential to deliver individualized risk assessments, equipping clinicians with the tools to refine both preventive and therapeutic interventions on a patient-by-patient basis. Moreover, advanced ensemble models like CatBoost [23] and LightGBM [24], particularly when enhanced through advanced metaheuristic optimization, remain largely underexplored for their role in cardiovascular disease risk estimation. Finally, the advanced two-tier frameworks such as the one proposed in this manuscript were not evaluated for this particular problem.

2.1. Machine and Deep Learning Models

This section provides a concise overview of the key machine learning and deep learning models employed in this study, namely CNN, CatBoost, and LightGBM. Each of these methods represents a distinct paradigm within modern predictive modeling: CNNs as powerful DL architectures tailored for extracting hierarchical representations from structured data, and CatBoost and LightGBM as advanced gradient boosting classifiers recognized for handling tabular datasets with high efficiency and accuracy. By briefly outlining their underlying principles and unique advantages, this section sets the foundation for understanding their role and relevance in the context of the present research.

CNNs [40] are a specialized class of deep learning architectures primarily designed for processing data with spatial or grid-like structures, including images or time-series signals. At their core, CNNs employ convolutional layers that use learnable filters (or kernels) to automatically identify and extract hierarchical features from raw input data, ranging from low-level edges and textures to complex patterns and objects in deeper layers. This hierarchical feature extraction reduces the need for manual feature engineering considerably and allows CNNs to generalize well across diverse tasks. Pooling layers are often incorporated to downsample feature maps, improving computational efficiency and reducing sensitivity to noise or local variations. Combined with fully connected layers and nonlinear activation functions, CNNs achieve strong performance in classification, detection, and segmentation tasks. Beyond computer vision, CNNs have also been successfully applied to fields such as natural language processing, bioinformatics, and medical imaging, where structured patterns and local dependencies are critical [41–43]. Their scalability, robustness, and ability to capture high-dimensional representations make them one of the most influential models in modern machine learning.

CatBoost (Categorical Boosting) is a gradient boosting library developed by Yandex that is particularly notable for its efficient and effective handling of categorical features [23]. Unlike many other boosting frameworks that require extensive preprocessing, CatBoost incorporates built-in algorithms to convert categorical variables into numerical representations in a way that reduces target leakage and preserves predictive power. The model is based on the principle of ordered boosting, a technique designed to eliminate prediction shift and overfitting that can occur when standard boosting methods use the same data both for training and target statistics calculation. CatBoost also supports symmetric or oblivious decision trees, which impose structural constraints that improve generalization and inference speed. These innovations make CatBoost highly competitive in scenarios

involving structured, tabular data with mixed feature types, such as financial modeling, recommendation systems, and medical diagnostics [44,45]. Its ease of use, strong out-of-the-box performance, and reduced need for feature engineering have positioned CatBoost as a robust alternative to more traditional gradient boosting methods.

LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework developed by Microsoft that is optimized for both speed and efficiency, making it highly suitable for large-scale and high-dimensional datasets [24]. At its core, LightGBM introduces two key innovations: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). GOSS selectively retains instances with larger gradients while downsampling those with smaller gradients, ensuring that the most informative data points are prioritized during training without sacrificing accuracy. EFB, on the other hand, reduces dimensionality by combining mutually exclusive features into single bundles, thereby lowering computational complexity. Additionally, LightGBM employs a leaf-wise tree growth strategy with depth constraints, which often yields deeper, more specialized trees compared to traditional level-wise growth, resulting in higher accuracy on many tasks [46–48]. Known for its scalability, parallelization capabilities, and efficiency with sparse data, LightGBM has become a go-to choice for tasks including ranking, classification, and regression in competitive ML challenges. Its balance of performance, speed, and memory efficiency makes it especially valuable for real-world applications that require rapid iteration and deployment.

2.2. Metaheuristics Optimization

Metaheuristic algorithms are population-based optimization methods designed to efficiently explore complex, non-convex search spaces, making them well suited for hyperparameter tuning in high-dimensional ML models, which is a central challenge in ML. This process exerts a profound influence on model accuracy but is notorious for its high computational demands. According to the NFL theorem [20], no universal optimization technique can outperform all others across every possible task, which has fueled growing interest in metaheuristic algorithms. These strategies are well-suited for navigating complex, high-dimensional search realms and mitigating the risk of early stagnation during the search process. Methods like genetic algorithm (GA) [49] and particle swarm optimization (PSO) [50] have demonstrated considerable effectiveness across a spectrum of fields, ranging from software engineering [21,51] and healthcare applications [52–54] to other computationally intensive domains [55–59]. Yet, within cardiovascular risk modeling, the application of metaheuristic-driven hyperparameter tuning for ML and DL models remains insufficiently explored. In particular, the VNS [25] stands out as an underutilized approach, whether employed in its original design or in modified, problem-specific variants.

3. Methods

Within this section, a baseline implementation of VNS is provided first, followed by the proposed adapted version later used in the simulations to tune the models of the proposed framework. Lastly, this section also outlines the architecture of the suggested two-tier framework for CVR classification.

3.1. Original Variable Neighborhood Search

The fundamental principle of the VNS [25] algorithm lies in its deliberate alteration of neighborhood structures throughout the optimization process. By doing so, VNS achieves a broader and more efficient traversal of the solution realm, which is especially valuable for tackling complex optimization problems. Unlike traditional approaches that restrict the search to a single, fixed neighborhood, VNS adaptively alternates among multiple

neighborhoods. This dynamic mechanism enables the method to escape local optima more effectively, thereby raising the chances of identifying superior solutions and ultimately improving algorithmic performance.

The functioning of VNS unfolds in three sequential phases: shaking, local search, and neighborhood change. These stages are carefully orchestrated to maintain a balance between exploration (searching new and diverse regions) and exploitation (intensifying the search near promising solutions). In the shaking phase, random modifications are introduced to the current solution within the neighborhood under consideration. This perturbation serves as a diversification tool, preventing premature convergence and helping the search move toward potentially better regions of the solution space.

Formally, within the k-th neighborhood, a new candidate solution x' is selected from the set $N_k(x)$, where the parameter k specifies the structure or extent of the neighborhood. This process is mathematically expressed in Equation (1).

$$x' \in N_k(x). \tag{1}$$

Here, the $N_k(x)$ denotes the adapted neighborhood where the solution x' is synthesized.

Once a new candidate solution is generated in the shaking phase, the algorithm proceeds to refine it through a local search. This search is carried out for x'' within the same neighborhood structure, aiming to minimize the objective function $f(x)$ in accordance with Equation (2).

$$f(x'') \leq f(x'). \tag{2}$$

During this stage, the algorithm conducts a focused exploration within the current neighborhood in search of possible refinements. A variety of local search strategies may be applied here, enabling the VNS procedure to adapt to the unique properties of the optimization problem. When the process uncovers a superior solution, more precisely, if the condition $f(x'') < f(x)$ holds, the incumbent solution x is replaced with the improved candidate x'', and the neighborhood counter k is reset to 1 to restart the search with renewed focus. If, however, no better solution emerges, the method advances to the subsequent neighborhood structure, as described in Equation (3).

$$k \leftarrow k + 1. \tag{3}$$

The VNS methodology establishes a flexible mechanism that continuously negotiates the trade-off between exploration and exploitation. The search unfolds in an iterative manner and persists until a stopping criterion is satisfied, for example hitting a maximum iteration limit or meeting a convergence requirement. What sets VNS apart is its inherent versatility, derived from its ability to cover multiple neighborhood structures rather than relying on a single one. This adaptability strengthens the algorithm's resilience, greatly minimizing the risk of premature entrapment in local optima and boosting its overall efficiency when applied to a wide range of optimization problems.

3.2. Modified VNS

Traditional VNS remains a powerful optimization tool and is still widely applied across many contemporary problem domains. Nevertheless, evaluations on modern benchmark functions have highlighted certain limitations. In particular, VNS often struggles with insufficient exploration in the early stages of the search and exhibits a strong inclination toward premature convergence, both of which restrict its ability to comprehensively investigate the solution space.

The conventional VNS often experiences a decline in exploratory power as the search progresses, since diversification naturally diminishes in later iterations. To address this

limitation, the modified VNS introduces a mechanism designed to regulate population diversity. In the early stages, maintaining high diversity is beneficial to broaden the search, while in later stages exploration should give way to more focused exploitation. The enhanced approach therefore adjusts diversity dynamically, encouraging wide exploration initially and gradually shifting toward intensification as the search advances.

Population diversity is quantified through the L_1-based diversity measure [60], which measures the spread of candidate solutions across the decision space. With N denoting the population size and D the problem's dimensionality, the L_1-based diversity measure is computed according to Equations (4)–(6).

$$\overline{x_j} = \frac{1}{N}\sum_{i=1}^{N} x_{ij}, \tag{4}$$

$$Dv_j^p = \frac{1}{N}\sum_{i=1}^{N} |x_{ij} - \overline{x}j|, \tag{5}$$

$$Dv^p = \frac{1}{D}\sum j = 1^D Dv_j^p. \tag{6}$$

In this formulation, $\overline{x}_j$ denotes the mean position of all individuals along dimension j, while Dv_j^p reflects the diversity contribution specific to that dimension. The diversity measure defined in Equations (4)–(6) is based on coordinate-wise absolute deviations across the population. While this formulation is related to the L1 norm through its use of absolute differences, it more closely corresponds to a mean absolute deviation computed per dimension and subsequently aggregated across all decision variables. In this work, the term L1-based is used to emphasize the reliance on absolute distances rather than squared deviations.

This diversity metric was selected due to its suitability for mixed continuous-integer hyperparameter spaces and its robustness to extreme values. Unlike variance-based or Euclidean-distance measures, the use of absolute deviations ensures stable behavior in the presence of heterogeneous parameter scales and avoids overemphasizing large deviations in individual dimensions. Moreover, the metric is computationally efficient and provides an interpretable measure of population spread, making it well suited for adaptive diversity control within the VNS framework.

At the outset of the search, population diversity is typically high but tends to decline progressively over time. To regulate this process, the modified VNS employs a dynamic threshold Dv_t that governs acceptable diversity levels. Whenever the observed diversity Dv_p falls below Dv_t during an iteration, a fraction of the population is replaced with newly generated candidates. To implement this mechanism, an auxiliary parameter NRS is introduced, specifying how many solutions are substituted at each update. Empirical studies indicate that the most effective results are obtained with $NRS = 2$. This replacement size parameter NRS was consequently fixed to 2 in this study. Preliminary experiments were conducted with alternative values of NRS relative to the population size ($NRS = 1$ and $NRS = 3$), corresponding to replacing 12.5% and 37.5% of the population, respectively. These configurations consistently resulted in inferior convergence behavior and reduced classification performance compared to $NRS = 2$. Replacing a single individual was insufficient to restore population diversity when stagnation occurred, while larger replacement rates introduced excessive randomness and hindered exploitation. The selected value $NRS = 2$ therefore represents a balanced trade-off between exploration and exploitation for the considered population size. The initial threshold Dv_{t0} is defined according to Equation (7),

$$Dv_{t0} = \sum_{j=1}^{NP} \frac{(ub_j - lb_j)}{2 \cdot NP}, \tag{7}$$

and recomputed during each round pertaining to Equation (8).

$$Dv_{t+1} = Dv_t - Dv_t \cdot \frac{t}{T}. \tag{8}$$

Here, t refers to the current round, while T designates the maximum number of iterations. NP denotes the population size, while ub_j and lb_j represent the upper and lower bounds of the j-th variable, respectively. The term $(ub_j - lb_j)$ therefore reflects the allowable search range along each dimension. As a result, the threshold Dv_t decreases over time, gradually shifting the algorithm's emphasis from broad exploration toward more intensive exploitation. The linear decay rule in Equation (8) was selected for its simplicity and predictable behavior, allowing a gradual and controlled transition from exploration to exploitation. Compared to exponential or nonlinear decay schemes, linear decay avoids abrupt reductions in diversity during early iterations, which could otherwise lead to premature convergence. Since $0 \leq \frac{t}{T} \leq 1$, the update rule in Equation (8) ensures that $Dv_{t+1} \geq 0$ for all iterations, consequently guaranteeing non-negativity of the diversity threshold. This adjustment is computationally efficient, introducing no significant overhead in terms of either processing time or memory usage.

The modified VNS is labeled adaptive population diversity aware VNS (APDAVNS) to reflect the incorporated changes. The pseudocode for the suggested APDAVNS is outlined in Algorithm 1.

Algorithm 1 APDAVNS pseudocode.

```
Synthesize initial swarm P
Compute Div_t0 and Div_t
while t < T do
  Execute original VNS search
  Compute fitness scores of solutions within P
  if Div_t < Div_p then
    Replace the NRS poorest solutions in P with novel individuals
  end if
  Compute Div_t
end while
return The finest solution.
```

To assess the computational overhead introduced by the adaptive population diversity mechanism, the average runtime of the original VNS and the proposed APDAVNS was compared under identical experimental settings. Since the diversity calculation involves only simple arithmetic operations and the replacement strategy affects a fixed number of individuals ($NRS = 2$), the number of fitness function evaluations (FFEs) remains unchanged. Since FFE is the most computationally expensive operation in the entire algorithm's run, the complexity of APDAVNS is identical to VNS in terms of FFEs. Empirical results confirmed that the additional operations introduce only a negligible runtime increase (below 2%), indicating that the proposed modification does not impose significant computational overhead.

3.3. Proposed Two-Tier Framework for CVR Classification

The proposed simulation framework follows a two-level hierarchical design, denoted as L1 and L2, and is based on a staged metaheuristic optimization strategy. In the first level (L1), a CNN is trained and optimized to perform feature extraction and dimensionality reduction, thereby transforming the original input space into a compact, high-level representation. Metaheuristic optimization is applied exclusively at this stage to tune the CNN

hyperparameters and identify the best-performing CNN network configuration. Once this optimal CNN is obtained, its parameters are fixed and the network is frozen.

In the second level (L2), the frozen CNN is truncated at a final dense layer and used solely as a feature extractor. The resulting fixed-length embedding vectors are then supplied as input to CatBoost and LightGBM classifiers. At this stage, no further training of the CNN is performed. Instead, metaheuristic optimization is applied only to the hyperparameters of the boosting classifiers to enhance classification performance. This staged optimization scheme decouples deep feature learning from ensemble-based classification, enabling efficient training while preserving model reproducibility. A schematic overview of the complete two-tier framework is illustrated in Figure 1.

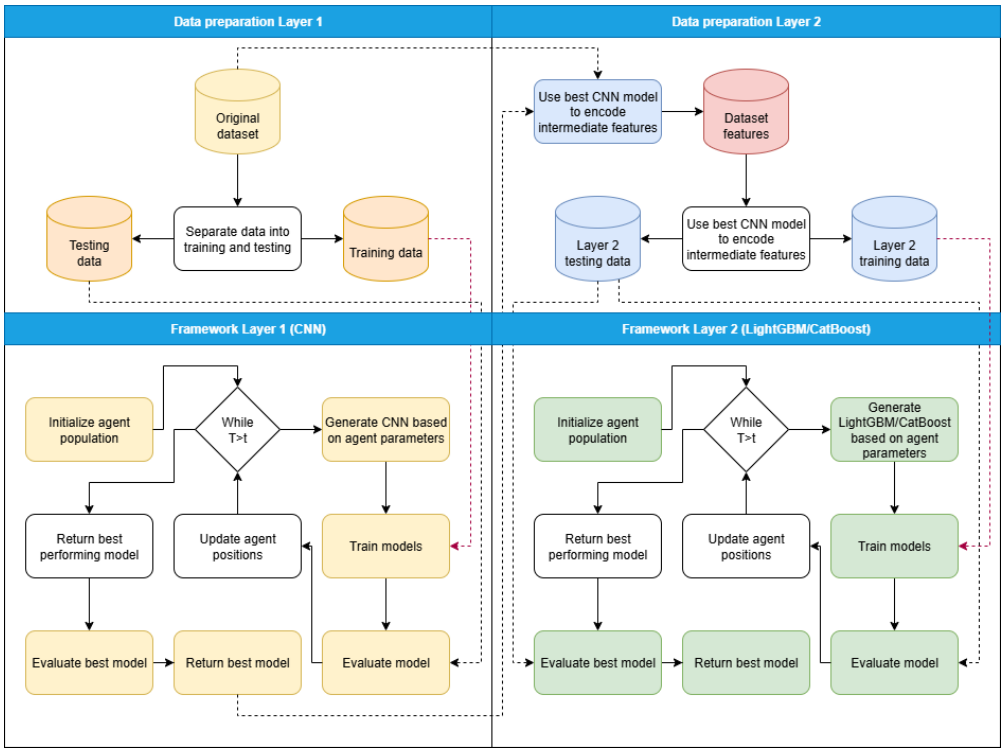


Figure 1. Overview of the proposed two-level framework. In L1, the CNN is trained and optimized using metaheuristic search. The best-performing CNN is then frozen and truncated to extract fixed-length feature embeddings, which are used as input for CatBoost and LightGBM optimization in L2.

It is important to note that the train–test split is performed once at the beginning of the pipeline using stratified sampling, and the test set is never used during model training at any level. All optimization steps in L1 and L2 are carried out by training models exclusively on the training data, while the test set is used only for final performance evaluation of the models.

Additionally, the proposed two-tier framework does not depend exclusively on the APDAVNS optimizer to function. Classical VNS can be used as a baseline optimization strategy. However, the experimental results demonstrate that APDAVNS provides more stable convergence and consistently superior performance across multiple classifiers and runs. The role of APDAVNS in this work is therefore not to enable the framework per se, but to enhance the efficiency and robustness of hyperparameter optimization in a high-dimensional, mixed search space.

4. Dataset Description and Simulation Setup

4.1. Exploratory Data Analysis

This study makes use of a publicly accessible cardiovascular risk dataset https://www.kaggle.com/datasets/alphiree/cardiovascular-diseases-risk-prediction-dataset [61]

(accessed on 15 November 2025). The dataset comprises 19 attributes encompassing demographic, lifestyle, and health-related factors. Demographic variables include age category and biological sex. Lifestyle-related features consist of smoking history, alcohol consumption, fruit and vegetable intake, and fried food consumption. Anthropometric measurements are represented by body mass index (BMI), height, and weight. Health status variables include self-reported general health, recent medical checkups, and the presence of selected comorbid conditions such as diabetes mellitus, arthritis, depression, and cancer. All comorbidity indicators are encoded as binary variables reflecting self-reported diagnoses. The target variable denotes the presence or absence of cardiovascular disease risk and is treated as a binary classification label. A key difficulty arises from the dataset's pronounced class imbalance, which complicates the ability of learning algorithms to reliably identify the minority class.

It should be noted that the present study does not address a time-to-event or survival analysis problem. Instead, the task is formulated as a binary classification problem based on cross-sectional data, where the objective is to predict whether an individual is classified as having CVD risk at the time of data collection. No explicit temporal horizon or future event window is modeled, and the outcome variable reflects current risk status rather than time-dependent incidence.

The preprocessing stage converted categorical features into numerical representations, employing one-hot encoding for certain variables and ordinal encoding for others. For conciseness, the specific encoding scheme is not elaborated here. The class distribution of the utilized dataset is illustrated in Figure 2. The feature correlation heatmap is illustrated in Figure 3. For model development, 70% of the data was reserved for training, with the remaining 30% allocated to evaluate predictive performance.

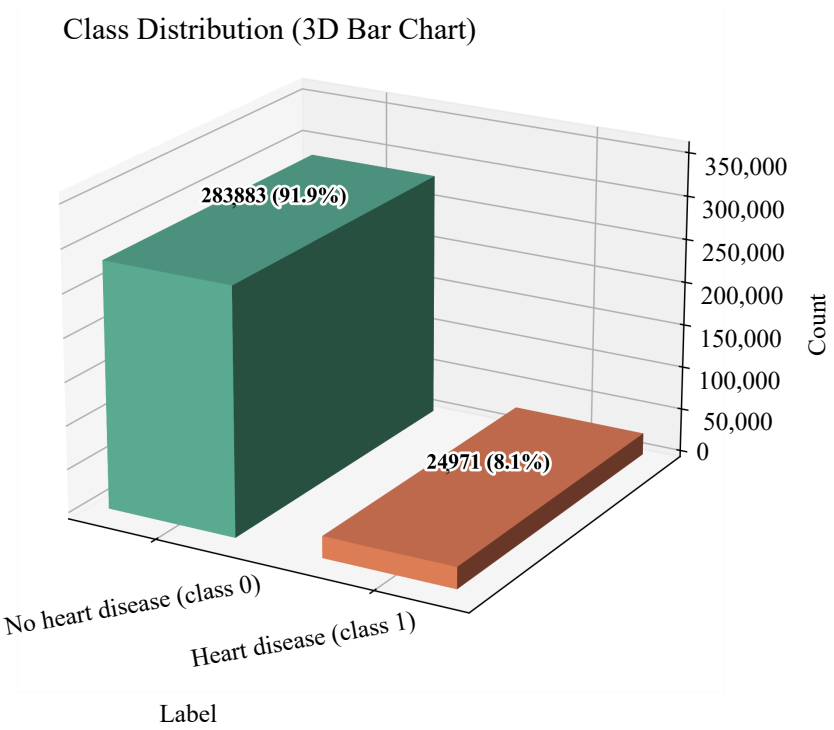


Figure 2. Distribution of classes of cardiovascular risk prediction dataset.

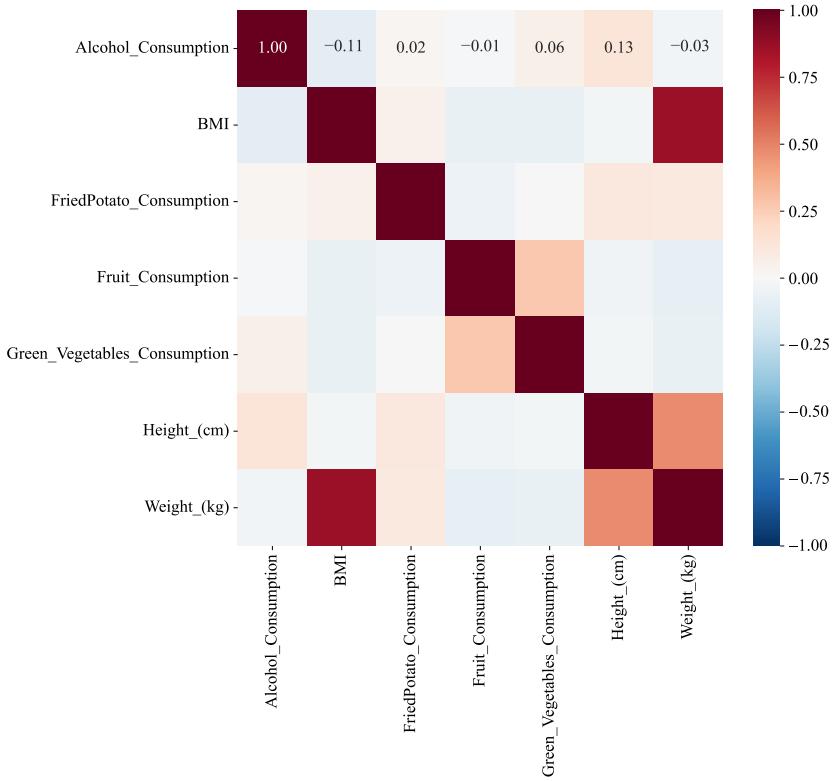


Figure 3. Correlation heatmap of the attributes within the employed CVD dataset.

The dataset was partitioned into training and testing subsets using a stratified 70/30 split to preserve the original class distribution in both subsets. To ensure reproducibility and consistency across all experiments, the same fixed random seed was used for data splitting and model training. This procedure guarantees that each class is proportionally represented in all subsamples and that reported results are directly comparable across different optimization strategies.

In addition to the reported hold-out evaluation, preliminary experiments were conducted using stratified cross-validation to assess the stability of the results across multiple splits. These experiments yielded performance trends consistent with those obtained using the fixed 70/30 split and did not lead to improved results. Given the already extensive experimental scope of the study, only the hold-out evaluation results are reported for clarity and conciseness.

A potential limitation of the employed dataset is the risk of information leakage, which is inherent to cross-sectional, self-reported health data. Certain variables, such as general health status or frequency of medical checkups, may partially reflect healthcare utilization patterns that occur after the onset of cardiovascular conditions rather than antecedent risk factors. To mitigate this risk, no features directly encoding cardiovascular diagnoses or treatments were included, and all variables were treated as static inputs without temporal ordering. Nevertheless, residual proxy leakage cannot be entirely excluded and may contribute to optimistic performance estimates. This limitation underscores that the proposed framework should be interpreted as a proof-of-concept for algorithmic risk modeling rather than a fully validated clinical decision-support system.

4.2. Evaluation Metrics

During the experimental evaluation, the effectiveness of the models was assessed through commonly used classification metrics, encompassing accuracy, precision, recall, and the F1 score, as formally specified in Equations (9)–(12).

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}, \tag{9}$$

$$Precision = \frac{TP}{TP + FP}, \tag{10}$$

$$Recall = \frac{TP}{TP + FN}, \tag{11}$$

$$F1_score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}, \tag{12}$$

here, TP, TN, FP, and FN mark the number of true positives, true negatives, false positives, and false negatives, respectively.

Beyond the standard classification metrics, the Matthews correlation coefficient (MCC) [62] was also adopted as performance measure. MCC is especially effective for imbalanced classification tasks, making it highly relevant in this study given the dataset's skewed class distribution and the presence of two minority classes. The coefficient is calculated according to Equation (13).

$$\text{MCC} = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \tag{13}$$

Across all simulation runs, the MCC served as the objective function, with the optimization process directed toward maximizing its value.

4.3. Experimental Setup

Three experimental scenarios were designed, each employing metaheuristic optimizers to adjust parameters across the two levels of the proposed dual-layer framework. The first tier comprised a CNN dedicated to feature extraction, while the second tier integrated CatBoost and LightGBM as classification models. In all cases, the custom CDGPSO algorithm was adopted as the primary optimizer, and its effectiveness was systematically benchmarked against a diverse collection of established optimization methods. This comparison set included the classical VNS [25], genetic algorithm (GA) [49], particle swarm optimization (PSO) [50], firefly algorithm (FA) [63], whale optimization algorithm (WOA) [64], sinh cosh optimizer (SCHO) [65] and COLSHADE [66], ensuring a representative balance between traditional and modern approaches. Each algorithm was implemented using the parameter configurations prescribed in the original references. To maintain uniformity across all experiments, every optimizer was run with a population size of $N = 8$ solutions. To mitigate stochastic variability, each configuration was repeated independently over 30 runs. For L1 optimization, each run consisted of $max_iter = 6$ rounds, while for L2 experiments, each run consisted of $max_iter = 10$ iterations. A smaller number of iterations was allocated to the L1 stage, given that CNNs are substantially more resource-intensive and computationally demanding than the models employed in L2. As outlined earlier, the optimization objective was the maximization of the Matthews correlation coefficient (MCC).

In the first experimental setting, the metaheuristic methods were tasked with tuning the CNN hyperparameters in the feature extraction layer. The parameters subjected to optimization are listed in Table 1, with an emphasis on compact and efficient architectures. The batch size was fixed at 256, and early stopping was applied after one-third of the designated epochs ($early\ stop = epochs/3$). Within the cooperative scheme, the optimized CNN obtained in this phase constituted the foundation for subsequent training and optimization in the second tier of the architecture.

Table 1. Tuned CNN hyperparameters with respective search ranges.

Hyperparameter	Low	High
Learning rate	0.0001	0.003
Dropout	0.05	0.2
Epochs	10	20
Convolutional layers	1	2
Dense layers	1	2
Cells per layer	16	32

The second experimental setup focused on the CatBoost classifier integrated into the second tier of the proposed framework. Using the feature representations previously derived from the CNN, CatBoost was subjected to hyperparameter optimization. The chosen parameters and their associated search intervals are summarized in Table 2.

Table 2. Tuned CatBoost hyperparameters with respective search ranges.

Hyperparameter	Low	High
Learning Rate	0.001	0.1
Col sample by level (cbl)	0.05	1
Subsample	0.05	1
Iterations	50	100
Depth	1	10
Minimum data in leaf (mdl)	5	20

The third experimental study concentrated on deploying LightGBM in the classification tier of the framework. For this purpose, feature embeddings were derived from the top-performing CNN identified in the initial stage, specifically captured after the dropout layer for each data instance. The transformed features subsequently served as input for training and hyperparameter tuning of the LightGBM classifier. The optimized hyperparameters considered, together with their respective search spaces, are reported in Table 3.

Table 3. Tuned LightGBM hyperparameters with respective search ranges.

Hyperparameter	Low	High
number of rounds	1	20
max depth	5	10
number of leaves	5	20
min child weight	1	5
feature fraction	0.1	0.9
bagging fraction	0.5	1
min split gain	0.001	0.1
λ L1 regularization	0	5
λ L2 regularization	0	3
learning rate	0.01	0.9

4.4. Solution Encoding, Neighborhood Definition and Shaking Phase

In all optimization experiments, candidate solutions are represented using a flat vector encoding. Each solution corresponds to a single individual in the population, where each vector component maps directly to one hyperparameter of the model being tuned. The dimensionality of the solution vector is therefore equal to the number of hyperparameters under optimization in the given experiment. Continuous hyperparameters (like learning rate, dropout rate, regularization coefficients) are encoded as real-valued variables within predefined bounds. Integer-valued hyperparameters (like number of layers, number of

neurons, tree depth, number of iterations) are encoded as continuous values during the search and discretized via rounding prior to model evaluation.

In the CNN optimization experiments, each candidate solution is encoded as a vector of length $d = 9$, with individual components representing the learning rate, dropout rate, number of training epochs, number of convolutional layers, number of dense layers, and the number of neurons assigned to each of the potential convolutional and dense layers (up to two layers of each type), with bounds specified in Table 1. In the CatBoost optimization simulations, each solution is encoded as a vector of length $d = 6$, with each parameter corresponding to one optimized parameter described in Table 2, with search limits given within the same table. Lastly, for the LightGBM simulations, each candidate solution is encoded as a vector of length $d = 10$, with each parameter corresponding to one optimized parameter described in Table 3, with search limits for each parameter given within the same table.

4.4.1. Neighborhood Structures

Neighborhood structures in the VNS framework are defined in a parameter-wise manner [25]. For continuous parameters, neighborhood moves are generated by applying bounded random perturbations within the parameter range. For integer parameters, neighborhood moves consist of incremental or discrete changes to adjacent admissible values. Boundary constraints are enforced after each neighborhood move to ensure that all candidate solutions remain within the predefined parameter ranges. This type-aware neighborhood design ensures that all generated candidate solutions remain valid and that the search process can effectively explore mixed continuous-discrete hyperparameter spaces.

4.4.2. Shaking Phase

In the shaking phase of the VNS algorithm, new candidate solutions are generated by applying random perturbations to the current solution within the active neighborhood structure. For continuous parameters, perturbations are sampled from a uniform distribution bounded by the predefined parameter limits. Integer-valued parameters are perturbed in a discrete manner and subsequently rounded to the nearest admissible integer value.

To ensure feasibility, all candidate solutions generated during the shaking phase are subjected to constraint handling before evaluation. Specifically, parameter values exceeding their predefined bounds are projected back into the feasible range, and discretization is applied where necessary. This procedure guarantees that all shaken solutions satisfy the parameter constraints and remain valid candidates for evaluation. This shaking strategy follows the standard VNS configuration commonly adopted in hyperparameter optimization, without introducing problem-specific bias [25].

4.4.3. Local Search Strategy and Stopping Criteria

In this work, the classical VNS strategy is employed without problem-specific modification [25]. Following the shaking phase, a local search is applied to the perturbed solution using a greedy improvement rule, whereby parameter-wise adjustments are evaluated sequentially and a new solution is accepted immediately if it yields an improvement in the objective function. Neighborhoods are explored in increasing order of size (such as increasing perturbation magnitude), and whenever an improving solution is found, the search is restarted from the first neighborhood, in accordance with the standard VNS scheme.

The optimization process is terminated based on a predefined maximum number of iterations, as specified for each experimental setting. No additional convergence-based stopping criteria are applied, ensuring consistent computational budgets across all optimization runs.

5. Experimental Outcomes

The experimental design centered on employing CNN models within the initial layer of the framework to condense and enhance the feature representations. In the following layer, classification was carried out using gradient boosting techniques. Among the evaluated methods at this stage, CatBoost and LightGBM were employed, both of which yielded strong and stable performance across conducted experiments. All reported performance metrics are computed on a held-out test set that is not used during model training or optimization. Consequently, the reported results reflect generalization performance on unseen data rather than in-sample or training error. The variability reported across repeated runs corresponds to different optimization initializations while preserving the same train–test split.

5.1. CNN L1

Table 4 summarizes the performance of CNN classifiers tuned with different metaheuristic algorithms, with particular emphasis on the MCC as the optimization objective. Among the tested approaches, the proposed APDAVNS achieved the best outcome, attaining a peak MCC of 0.179496. Beyond excelling in the best-case result, APDAVNS also delivered leading values for the worst-case (0.150341), mean (0.169199), and median (0.173480), underscoring both stability and robustness. In contrast, the PSO algorithm stood out for its consistency, showing the lowest variability across repeated runs.

Table 4. Fitness values for L1 CNN tuning (best scores bolded).

Method	Best	Worst	Mean	Median	Std	Var
CNN-APDAVNS	**0.179496**	**0.150341**	**0.169199**	**0.173480**	0.011274	0.000127
CNN-VNS	0.159240	0.119970	0.137545	0.135485	0.014370	0.000206
CNN-GA	0.156340	0.138457	0.151314	0.155229	0.007463	5.57×10^{-5}
CNN-PSO	0.153203	0.140981	0.145508	0.143924	**0.004738**	$\mathbf{2.24 \times 10^{-5}}$
CNN-FA	0.152539	0.132665	0.143622	0.144642	0.007112	5.06×10^{-5}
CNN-WOA	0.159184	0.129977	0.148184	0.151787	0.010938	0.000120
CNN-SCHO	0.165382	0.147324	0.153516	0.150679	0.006987	4.88×10^{-5}
CNN-COLSHADE	0.158768	0.145269	0.151373	0.150728	0.004852	2.35×10^{-5}

Table 5 presents a comparison of CNN classifiers optimized with various metaheuristic strategies, assessed using the indicator function (error rate). The best overall error rate of 0.080490 was recorded by the baseline PSO, with the proposed APDAVNS producing a slightly lower result. PSO showed superior reliability across other indicator metrics, achieving the lowest worst-case (0.080426), mean (0.080550), and median (0.080458) error rates, although it must be highlighted that error rate was not the target of the optimization. While APDAVNS did not secure the leading scores, it exhibited strong consistency across independent runs, reflecting stable though slightly less competitive outcomes regarding the indicator function.

Table 6 reports the full set of performance indicators for the top CNN classifiers obtained through each of the metaheuristic optimization methods. The findings highlight that the proposed APDAVNS produced a highly competitive CNN model, achieving an overall accuracy of 0.918376 and delivering strong precision, recall (sensitivity), and F1-scores across the evaluated classes. In this instance, the PSO-optimized CNN achieved the highest accuracy at 0.91951. A key observation, however, is that all CNN variants struggled with correct classification of the minority class (CVD risk), revealing a recurring limitation in addressing underrepresented outcomes. This shortcoming provides the motivation for the methodological improvements presented in the following sections.

Table 5. Indicator values for L1 CNN tuning.

Method	Best	Worst	Mean	Median	Std	Var
CNN-APDAVNS	0.081624	0.080695	0.081324	0.081402	0.000427	1.82×10^{-7}
CNN-VNS	0.080998	0.080544	0.080733	0.080695	0.000165	2.72×10^{-8}
CNN-GA	0.080814	0.080652	0.080917	0.080922	0.000198	3.94×10^{-8}
CNN-PSO	0.080490	0.080426	0.080550	0.080458	0.000269	7.24×10^{-8}
CNN-FA	0.080760	0.080598	0.080604	0.080663	0.000170	2.89×10^{-8}
CNN-WOA	0.080954	0.080480	0.080817	0.080917	0.000196	3.83×10^{-8}
CNN-SCHO	0.081160	0.081235	0.080917	0.080938	0.000288	8.27×10^{-8}
CNN-COLSHADE	0.080803	0.080663	0.080644	0.080722	0.000190	3.61×10^{-8}

Table 6. Detailed outcomes for the finest-produced CNN models.

Model	Metric	No Risk	Risk	Accuracy	Macro Ave	Weight Ave
CNN-APDAVNS	precision	0.925236	0.474468	0.918376	0.699852	0.888793
	recall	0.991299	0.089307	0.918376	0.540303	0.918376
	f1-score	0.957129	0.150320	0.918376	0.553725	0.891901
CNN-VNS	precision	0.919153	0	0.919153	0.459577	0.844843
	recall	1	0	0.919153	0.500000	0.919153
	f1-score	0.957874	0	0.919153	0.478937	0.880433
CNN-GA	precision	0.923470	0.501594	0.919186	0.712532	0.889363
	recall	0.994493	0.063009	0.919186	0.528751	0.919186
	f1-score	0.957667	0.111954	0.919186	0.534811	0.889294
CNN-PSO	precision	0.923126	0.519856	0.919510	0.721491	0.890523
	recall	0.995315	0.057669	0.919510	0.526492	0.919510
	f1-score	0.957863	0.103821	0.919510	0.530842	0.888816
CNN-FA	precision	0.923235	0.504525	0.919240	0.713880	0.889383
	recall	0.994857	0.059538	0.919240	0.527198	0.919240
	f1-score	0.957709	0.106507	0.919240	0.532108	0.888892
CNN-WOA	precision	0.919153	0	0.919153	0.459577	0.844843
	recall	1	0	0.919153	0.500000	0.919153
	f1-score	0.957874	0	0.919153	0.478937	0.880433
CNN-SCHO	precision	0.924156	0.487134	0.918840	0.705645	0.888824
	recall	0.993213	0.073288	0.918840	0.533251	0.918840
	f1-score	0.957441	0.127408	0.918840	0.542424	0.890336
CNN-COLSHADE	precision	0.923600	0.502066	0.919197	0.712833	0.889521
	recall	0.994340	0.064878	0.919197	0.529609	0.919197
	f1-score	0.957666	0.114907	0.919197	0.536287	0.889532
	support	85,166	7491			

To ensure reproducibility, the hyperparameter settings of the models that achieved the strongest outcomes are summarized in Table 7. Furthermore, Figure 4 depicts the structure of the top-performing CNN alongside its truncated version, which was employed to extract intermediate feature embeddings for the subsequent second-layer experiments with boosting classifiers.

Table 7. Determined parameters for the finest-performing L1 CNN structures.

Method	lr	Drop	Epochs	Dense-L	CNN-L	CNN1	CNN2	DL1	DL2
CNN-APDAVNS	0.002399	0.05	20	2	1	23	25	31	N/A
CNN-VNS	0.001887	0.138484	10	2	1	16	18	29	N/A
CNN-GA	0.003	0.05	10	2	2	32	32	20	16
CNN-PSO	0.003	0.05	10	1	1	30	N/A	25	N/A
CNN-FA	0.003	0.2	20	2	1	16	16	30	N/A

Table 7. *Cont.*

Method	lr	Drop	Epochs	Dense-L	CNN-L	CNN1	CNN2	DL1	DL2
CNN-WOA	0.002503	0.104784	12	2	1	30	21	28	N/A
CNN-SCHO	0.003	0.05	20	2	1	23	16	16	N/A
CNN-COLSHADE	0.003	0.123624	20	2	1	32	32	32	N/A

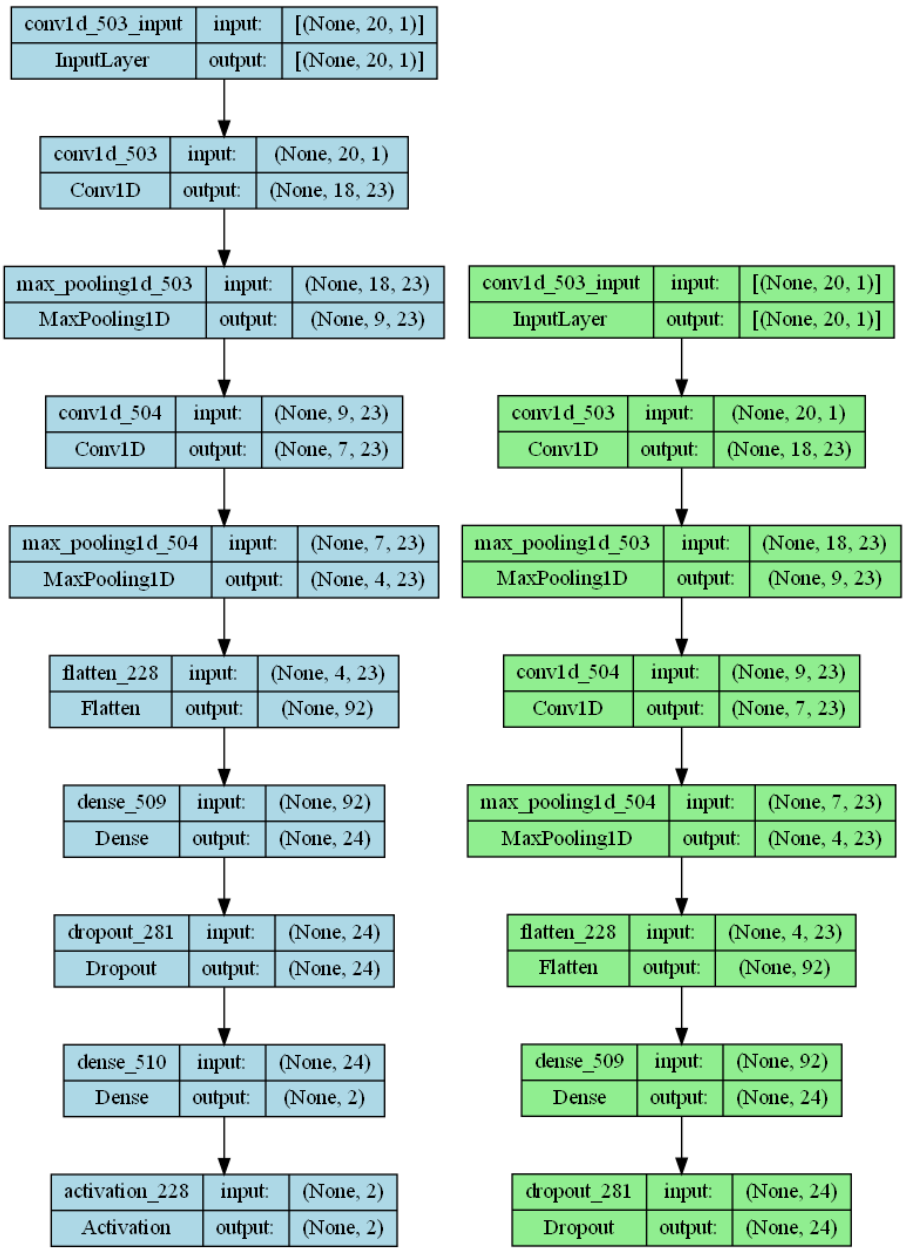


Figure 4. Finest-performing optimized CNN and truncated model architectures.

5.2. CNN + CatBoost L2

Table 8 summarizes the performance of CatBoost classifiers in the second layer, optimized with different metaheuristic algorithms and evaluated using the MCC as the fitness function. In this setting, the proposed APDAVNS emerged as the most effective optimizer, attaining the highest best-case MCC of 0.155012. Beyond this peak value, APDAVNS also secured leading results for the mean (0.150510) and median (0.150140) scores, underscoring both robustness and stability. FA obtained the best result for the worst-case run (0.148474). Lastly, WOA attained the lowest variability across repeated runs in this scenario.

Table 8. Fitness values for L2 CatBoost tuning (best scores bolded).

Method	Best	Worst	Mean	Median	Std	Var
CNN-CB-APDAVNS	**0.155012**	0.147137	**0.150510**	**0.150140**	0.002321	5.39×10^{-6}
CNN-CB-VNS	0.148474	0.139745	0.145916	0.147067	0.002890	8.35×10^{-6}
CNN-CB-GA	0.150296	0.142617	0.148206	0.148758	0.002346	5.51×10^{-6}
CNN-CB-PSO	0.152098	0.144766	0.147913	0.147821	0.002228	4.96×10^{-6}
CNN-CB-FA	0.153246	**0.148474**	0.149975	0.149643	0.001524	2.32×10^{-6}
CNN-CB-WOA	0.148815	0.144772	0.147735	0.148422	**0.001284**	$\mathbf{1.65 \times 10^{-6}}$
CNN-CB-SCHO	0.151133	0.142591	0.147773	0.147692	0.002350	5.52×10^{-6}
CNN-CB-COLSHADE	0.150954	0.143602	0.146893	0.147055	0.002828	8.00×10^{-6}

Table 9 presents the comparative results of CatBoost classifiers in the second layer, optimized with various metaheuristic strategies and assessed using the indicator function (error rate). FA outperformed competing methods, recording the lowest overall error rate of 0.079595. It also achieved the best mean value (0.079727). The proposed APDAVNS delivered the strongest worst-case performance with an error rate of 0.079713, and the best median score (0.079697). While SCHO did not match these absolute scores, it exhibited the highest stability across independent trials, reflecting reliable but slightly less competitive behavior.

Table 9. Indicator values for L2 CatBoost tuning.

Method	Best	Worst	Mean	Median	Std	Var
CNN-CB-APDAVNS	0.079670	0.079713	0.079740	0.079697	0.000139	1.94×10^{-8}
CNN-CB-VNS	0.079746	0.080070	0.079871	0.079843	0.000124	1.53×10^{-8}
CNN-CB-GA	0.079703	0.079962	0.079758	0.079719	0.000096	9.13×10^{-9}
CNN-CB-PSO	0.079659	0.079821	0.079819	0.079784	0.000138	1.90×10^{-8}
CNN-CB-FA	0.079595	0.079746	0.079727	0.079746	0.000074	5.53×10^{-9}
CNN-CB-WOA	0.079767	0.079994	0.079806	0.079784	0.000086	7.31×10^{-9}
CNN-CB-SCHO	0.079746	0.079940	0.079843	0.079832	0.000055	2.97×10^{-9}
CNN-CB-COLSHADE	0.079670	0.080113	0.079912	0.079902	0.000178	3.18×10^{-8}

Table 10 reports the comprehensive evaluation metrics for the top-performing CatBoost classifiers in the second layer, optimized through different metaheuristic algorithms. The findings indicate that the FA yielded the strongest model, reaching a classification accuracy of 0.920405. However, the proposed APDAVNS delivered the model with high precision, and better recall (sensitivity), and F1-scores across classes, particularly for minor class (CVD risk). An important observation is that observing accuracy only in this scenario is not enough, as the proposed CNN-CB-APDAVNS is the best in handling the minority class at the cost of slightly lower overall accuracy, which is vital in this scenario (detecting CVD risk). Another important observation is that all CatBoost-based models at the second layer achieved higher accuracy than standalone CNNs and showed improved recognition of minority class. Furthermore, CatBoost outperformed LightGBM when integrated into the same layer of the framework.

Table 10. L2 CNN-CB best generated models comprehensive assessment.

Approach	Metric	No Risk	Risk	Accuracy	Macro Ave	Weight Ave
CNN-CB-APDAVNS	precision	0.922702	0.583975	0.920330	0.753339	0.895317
	recall	0.996830	0.050594	0.920330	0.523712	0.920330
	f1-score	0.958335	0.093120	0.920330	0.525728	0.888385

Table 10. *Cont.*

Approach	Metric	No Risk	Risk	Accuracy	Macro Ave	Weight Ave
CNN-CB-VNS	precision	0.922393	0.586441	0.920254	0.754417	0.895233
	recall	0.997135	0.046189	0.920254	0.521662	0.920254
	f1-score	0.958309	0.085633	0.920254	0.521971	0.887756
CNN-CB-GA	precision	0.922461	0.588333	0.920297	0.755397	0.895448
	recall	0.997100	0.047123	0.920297	0.522111	0.920297
	f1-score	0.958329	0.087257	0.920297	0.522793	0.887906
CNN-CB-PSO	precision	0.922529	0.590164	0.920341	0.756346	0.895658
	recall	0.997065	0.048058	0.920341	0.522561	0.920341
	f1-score	0.958350	0.088878	0.920341	0.523614	0.888056
CNN-CB-FA	precision	0.922543	0.595710	0.920405	0.759126	0.896120
	recall	0.997123	0.048191	0.920405	0.522657	0.920405
	f1-score	0.958384	0.089169	0.920405	0.523777	0.888111
CNN-CB-WOA	precision	0.922428	0.583333	0.920233	0.752881	0.895014
	recall	0.997065	0.046723	0.920233	0.521894	0.920233
	f1-score	0.958295	0.086516	0.920233	0.522406	0.887815
CNN-CB-SCHO	precision	0.922540	0.581994	0.920254	0.752267	0.895008
	recall	0.996947	0.048325	0.920254	0.522636	0.920254
	f1-score	0.958302	0.089239	0.920254	0.523771	0.888041
CNN-CB-COLSHADE	precision	0.922473	0.590985	0.920330	0.756729	0.895673
	recall	0.997123	0.047257	0.920330	0.522190	0.920330
	f1-score	0.958346	0.087515	0.920330	0.522931	0.887943
	samples	85,166	7491			

Lastly, to facilitate experimental reproducibility, the hyperparameter selections of the best CatBoost models optimized by each algorithm that produced the best results within L2 of the suggested framework are delineated in Table 11.

Table 11. Best L2 CatBoost models determined by each algorithm.

Structure	lr	cbl	Subsample	Iterations	Depth	mdl
CNN-CB-APDAVNS	0.1	0.9379	0.9108	100	10	20
CNN-CB-VNS	0.1	1	1	98	10	5
CNN-CB-GA	0.1	0.7071	0.7553	98	10	19
CNN-CB-PSO	0.1	0.5965	0.05	100	10	7
CNN-CB-FA	0.1	0.8681	1	100	10	5
CNN-CB-WOA	0.1	0.5195	1	98	10	20
CNN-CB-SCHO	0.0806	0.9367	1	100	10	12
CNN-CB-COLSHADE	0.1	0.9162	1	100	10	5

5.3. CNN + LGBM L2

Table 12 compares the performance of LightGBM classifiers in the second layer, optimized with different metaheuristic techniques, using the MCC as the objective. The proposed APDAVNS achieved superior outcomes in this scenario, attaining the strongest best-case performance with a peak MCC of 0.296806 and also led in terms of worst (0.287135) and mean (0.291793), underscoring its robustness and reliability. On the other hand, the best median value was obtained by PSO in this scenario (0.291341). APDAVNS also exhibited the greatest stability across repeated trials, reflecting its consistency despite slightly lower overall scores.

Table 12. Fitness values for L2 LightGBM tuning (best scores bolded).

Method	Best	Worst	Mean	Median	Std	Var
CNN-LGBM-APDAVNS	**0.296806**	**0.287135**	**0.291793**	0.290558	**0.003288**	**1.08×10^{-5}**
CNN-LGBM-VNS	0.295877	0.183013	0.276185	0.289642	0.035992	0.001295
CNN-LGBM-GA	0.296335	0.192825	0.279607	0.290214	0.032947	0.001085
CNN-LGBM-PSO	0.295906	0.285908	0.291610	**0.291341**	0.003546	1.26×10^{-5}
CNN-LGBM-FA	0.296308	0.175875	0.276434	0.288871	0.038244	0.001463
CNN-LGBM-WOA	0.289532	0.165425	0.245381	0.288871	0.056403	0.003181
CNN-LGBM-SCHO	0.295906	0.271274	0.288380	0.289473	0.007153	5.12×10^{-5}
CNN-LGBM-COLSHADE	0.295953	0.191678	0.250601	0.271177	0.045990	0.002115

Table 13 summarizes the error rate outcomes of LightGBM classifiers in the second layer, optimized with different metaheuristic strategies. The lowest overall error of 0.101870 was achieved by the WOA algorithm. WOA also had the best worst result (0.079713) while COLSHADE proved most effective on average, producing the best mean (0.092392) and median (0.093857) error rates. While the proposed APDAVNS did not lead in absolute performance regarding the indicator function, it showed strong consistency across runs, underscoring its stability despite less competitive optimal values. It is also important to underpin once again that classification error rate was not the target for the optimization process, but only used as an indicator.

Table 13. Indicator values for L2 LightGBM tuning.

Method	Best	Worst	Mean	Median	Std	Var
CNN-LGBM-APDAVNS	0.105928	0.101126	0.103244	0.102000	0.002068	4.28×10^{-6}
CNN-LGBM-VNS	0.105907	0.080609	0.099429	0.101833	0.007922	6.28×10^{-5}
CNN-LGBM-GA	0.105864	0.080652	0.100053	0.101967	0.007471	5.58×10^{-5}
CNN-LGBM-PSO	0.104579	0.097435	0.102170	0.102151	0.002357	5.56×10^{-6}
CNN-LGBM-FA	0.105842	0.080210	0.099339	0.101417	0.007631	5.82×10^{-5}
CNN-LGBM-WOA	0.101870	0.079713	0.093506	0.101525	0.010443	0.000109
CNN-LGBM-SCHO	0.104579	0.094164	0.101308	0.101676	0.003038	9.23×10^{-6}
CNN-LGBM-COLSHADE	0.106069	0.080674	0.092392	0.093857	0.009866	9.73×10^{-5}

Table 14 reports the complete evaluation metrics for the best-performing LightGBM classifiers in the second layer, optimized by the different metaheuristic algorithms considered. The findings highlight that the WOA produced one of the strongest models, attaining the highest classification accuracy of 0.89813. However, CNN-LGBM-APDAVNS managed to deliver better precision, recall (sensitivity), and F1-scores across classes, particularly for the minor class (CVD risk). An important observation is that all LightGBM-based models at this layer did not outperform standalone CNNs, and demonstrated lower effectiveness than models utilizing CatBoost in L2.

For reproducibility, the hyperparameter settings of the LightGBM models that achieved the best results in the second layer of the framework are summarized in Table 15.

Table 14. Comprehensive assessment of best generated L2 CNN-LGBM models.

Approach	Metric	No Risk	Risk	Accuracy	Macro Ave	Weight Ave
CNN-LGBM-APDAVNS	precision	0.943424	0.348934	0.894072	0.646179	0.895361
	recall	0.941197	0.358297	0.894072	0.649747	0.894072
	f1-score	0.942309	0.353553	0.894072	0.647931	0.894710

Table 14. *Cont.*

Approach	Metric	No Risk	Risk	Accuracy	Macro Ave	Weight Ave
CNN-LGBM-VNS	precision	0.943415	0.348985	0.894093	0.646200	0.895357
	recall	0.941232	0.358163	0.894093	0.649698	0.894093
	f1-score	0.942322	0.353515	0.894093	0.647918	0.894719
CNN-LGBM-GA	precision	0.943459	0.349324	0.894136	0.646392	0.895425
	recall	0.941232	0.358697	0.894136	0.649965	0.894136
	f1-score	0.942344	0.353948	0.894136	0.648146	0.894775
CNN-LGBM-PSO	precision	0.943059	0.353028	0.895421	0.648043	0.895357
	recall	0.943170	0.352556	0.895421	0.647863	0.895421
	f1-score	0.943114	0.352792	0.895421	0.647953	0.895389
CNN-LGBM-FA	precision	0.943450	0.349376	0.894158	0.646413	0.895421
	recall	0.941268	0.358564	0.894158	0.649916	0.894158
	f1-score	0.942358	0.353910	0.894158	0.648134	0.894784
CNN-LGBM-WOA	precision	0.941552	0.358963	0.898130	0.650257	0.894451
	recall	0.948019	0.330930	0.898130	0.639475	0.898130
	f1-score	0.944774	0.344377	0.898130	0.644576	0.896234
CNN-LGBM-SCHO	precision	0.943059	0.353028	0.895421	0.648043	0.895357
	recall	0.943170	0.352556	0.895421	0.647863	0.895421
	f1-score	0.943114	0.352792	0.895421	0.647953	0.895389
CNN-LGBM-COLSHADE	precision	0.943467	0.348542	0.893931	0.646005	0.895370
	recall	0.940986	0.358964	0.893931	0.649975	0.893931
	f1-score	0.942225	0.353676	0.893931	0.647951	0.894643
	samples	85,166	7491			

Table 15. Finest L2 LightGBM models determined by each algorithm.

Approach	Rounds	Max_Depth	Leaves	mcw	ff	bf	msg	λ L1	λ L2	lr
CNN-LGBM-APDAVNS	1	5	16	5	0.2975	0.8509	0.0607	2.5211	0.2652	0.866
CNN-LGBM-VNS	1	10	20	5	0.2686	1	0.1	5	0	0.9
CNN-LGBM-GA	1	7	19	4	0.2645	0.6366	0.0224	0	1.5377	0.9
CNN-LGBM-PSO	1	6	14	5	0.258	0.5	0.0438	0	0	0.9
CNN-LGBM-FA	1	10	20	5	0.2986	0.9871	0.1	0	1.3438	0.9
CNN-LGBM-WOA	1	10	20	2	0.1	0.5	0.1	0	0	0.9
CNN-LGBM-SCHO	1	5	16	1	0.2668	0.8593	0.0276	1.6794	0.5743	0.9
CNN-LGBM-COLSHADE	1	7	13	2	0.2869	0.6453	0.02428	3.1483	0	0.9

5.4. Visual Comparative Analysis

Figure 5 presents a comprehensive comparative analysis of various metaheuristic optimizers used to fine-tune both levels of the suggested framework for classifying CVD risk. The study consisted of three distinct experiments, namely L1 CNN optimization (top row), L2 CatBoost optimization (middle row) and L2 LightGBM optimization (bottom row). The combination of box plots visualizes the empirical distribution of MCC scores and error rates over 30 separate optimization runs, offering a statistically grounded perspective on metaheuristics performance. These plots reveal crucial aspects of each method, like median values, dispersion, and skewness, emphasizing the balance between search stability and exploratory behavior. Notably, the proposed APDAVNS algorithm consistently yields superior best-case MCC scores across all three optimization experiments, indicative of its potent global search capabilities. It also exhibits excellent distributions of the objective over all three simulations. However, its broader distribution in some instances also reflects

purposeful exploration of the solution space, which serves to reduce the risk of early convergence to suboptimal solutions.

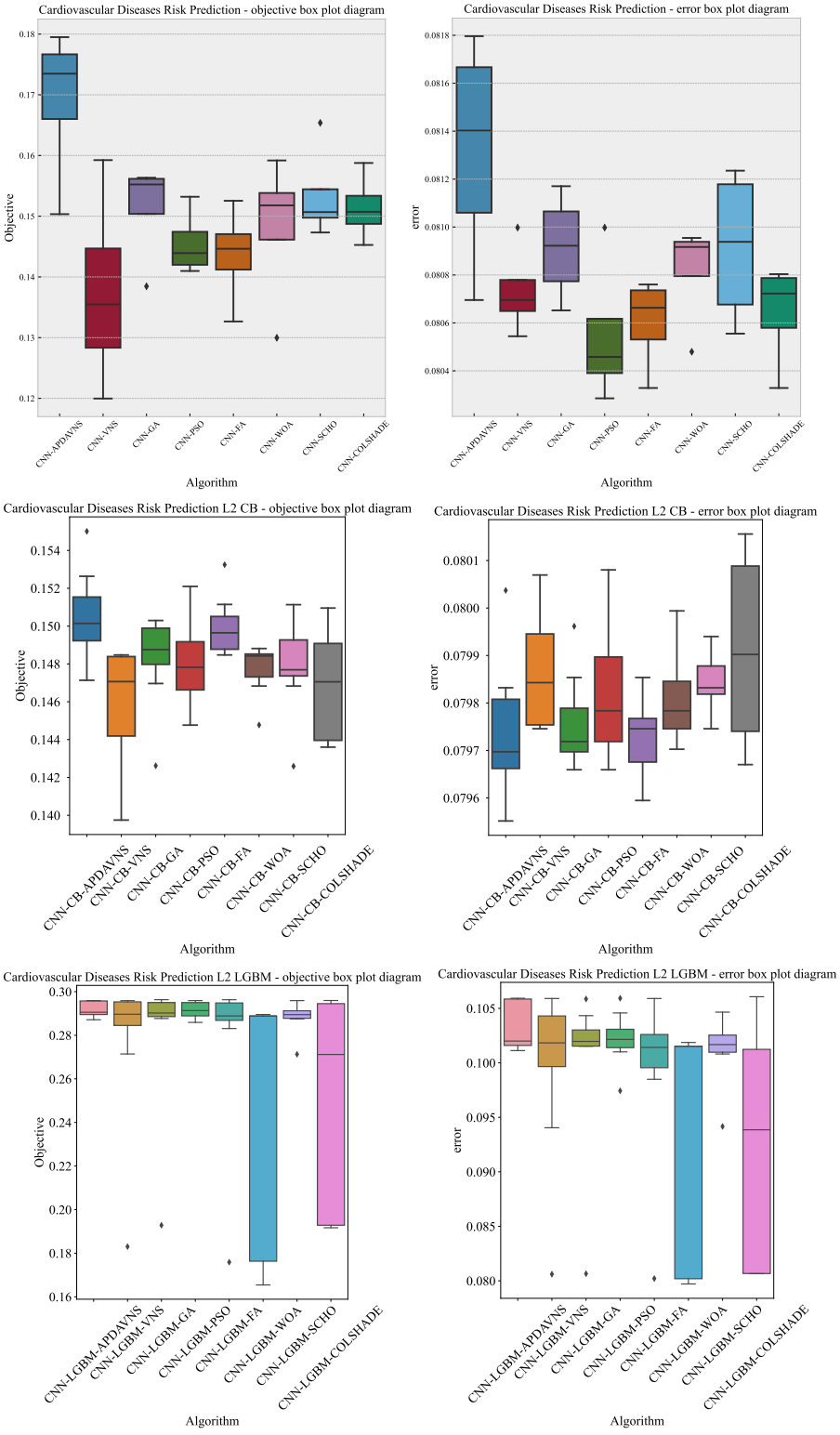


Figure 5. MCC and error dispersion for L1 CNN (**top**), L2 CatBoost (**middle**) and L2 LightGBM (**bottom**) experiments. Diamond symbols indicate outliers. Colors are used only for visual distinction.

This box plot analysis captures both the spread and the central tendency of error rates across 30 independent optimization executions, providing critical evidence of each

algorithm's capacity to generalize, particularly under class-imbalanced conditions. Methods characterized by lower median error rates and narrower interquartile ranges reflect greater reliability and solution quality across repeated stochastic executions. Among them, APDAVNS consistently performs well, attaining low error rates and demonstrating the strength of its adaptive strategy: maintaining diversity in exploration while effectively exploiting high-potential regions of the search space. This balance enhances resilience against entrapment in local optima and reduces the overall likelihood of misclassification. Again, it is important to emphasize that the classification error rate was not the primary target of optimization, as some of the models attained good overall accuracy with poor performance on the minority class (CVD risk).

The convergence curves shown in Figure 6 offer a temporal perspective on optimization behavior, illustrating how the objective function evolves across iterations for each algorithm. The proposed APDAVNS displays rapid and stable convergence, reflecting an effective balance between exploration and exploitation. Such performance can be attributed to its adaptive design, incorporating guided diversity to enhance search efficiency. By comparison, several competing algorithms exhibit slower improvement or premature stagnation, indicating weaker adaptability to the complex landscapes inherent in hyperparameter tuning.

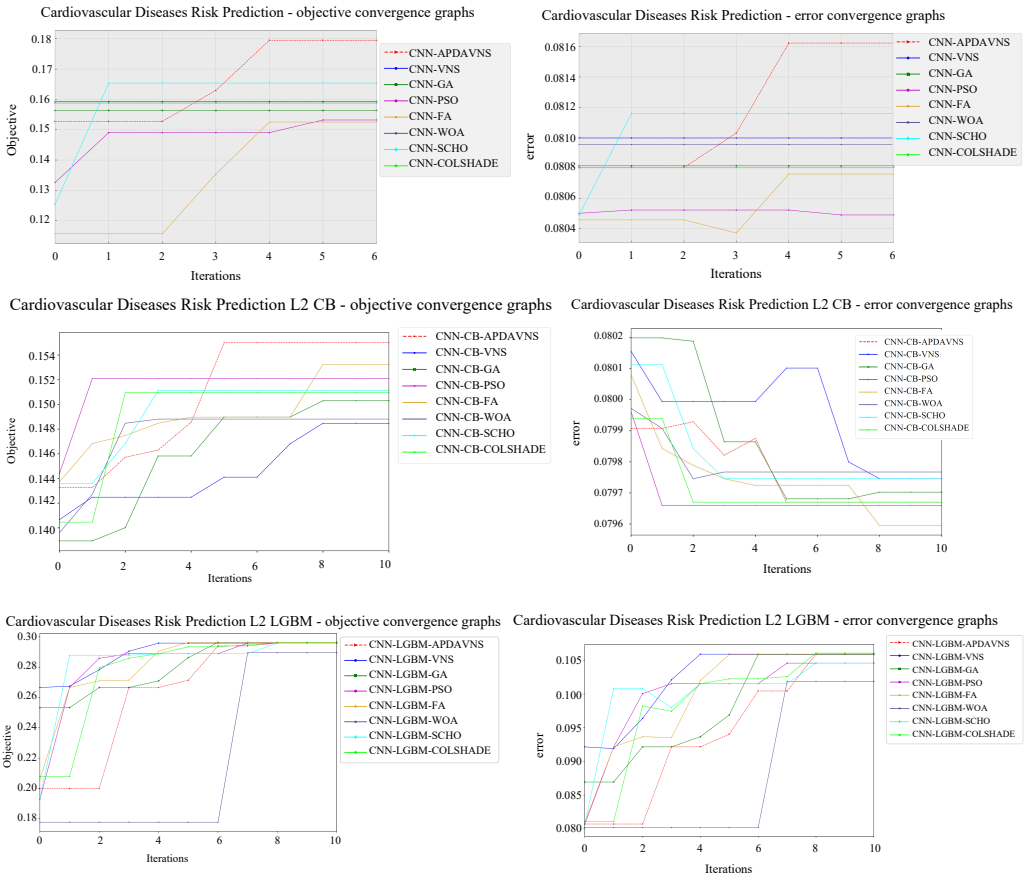


Figure 6. MCC and error convergence diagrams for L1 CNN (**top**), L2 CatBoost (**middle**) and L2 LightGBM (**bottom**) experiments.

The convergence graphs of the indicator function are also provided within Figure 6, showing how error rate evolves over the iterations. Again, algorithms that display rapid and stable convergence typically achieve a balanced shift between diversification and intensification, which is an essential property for effectively navigating high-dimensional

hyperparameter landscapes. Conversely, convergence curves that appear flat or irregular often signal limited adaptability or premature stagnation, stemming from inadequate exploration or excessively greedy exploitation. When considered alongside statistical analyses, these temporal patterns provide a richer perspective on the comparative strengths and shortcomings of each metaheuristic strategy.

Collectively, the visual analyses underscore how the underlying design of an optimization algorithm shapes its effectiveness in hyperparameter tuning for classification problems. Factors such as maintaining population diversity, adopting robust neighborhood search strategies, and incorporating adaptive diversity control directly affect convergence behavior, solution quality, and reproducibility. These aspects are particularly vital for practitioners aiming to implement reliable, high-performing models in real-world environments, where sensitivity to initialization and fluctuations in input data can substantially impact classification performance.

Figure 7 presents swarm plots illustrating the distribution and diversity of objective function values and error rates obtained by different metaheuristic algorithms at the final iteration of their respective best-performing run across three experimental settings: L1 CNN optimization (top row), L2 CatBoost optimization (middle row), and L2 LightGBM optimization (bottom row). These visualizations capture both the solution landscape and the quality of final outcomes. The proposed APDAVNS typically produces tightly clustered swarms around the best solutions, reflecting strong global exploration, reliable convergence, and population-level consistency.

By jointly displaying performance metrics and diversity, the plots reveal how effectively each optimizer manages the trade-off between exploitation and exploration, which is a crucial factor in preventing premature stagnation and uncovering generalized hyperparameter settings. This dual-perspective view is especially valuable in medical classification tasks, where reliability, adaptability, and robustness against initialization sensitivity are crucial for deploying trustworthy models under real-world data variability.

The radar diagrams of macro and weighted averages of the analyzed models are depicted within Figure 8. The macro averages reflect the model's ability to treat all classes equally, regardless of class size. This perspective is particularly important for highlighting weaknesses in detecting minority classes, since each class contributes equally to the overall score. The radar shape for the macro averages in this scenario indicates that the models perform in a balanced sense, even under severe class imbalance. The weighted averages, on the other hand, account for the relative frequency of each class in the dataset. These values emphasize performance on majority classes and in this case produce higher scores since the data distribution is highly skewed.

Evaluation of the fitness function, where larger values correspond to stronger MCC performance, indicates a clear hierarchy among the tested models: CatBoost at the L2 layer consistently surpasses LightGBM at L2 and also significantly outperforms the standalone CNN at L1. The box plots reveal that CatBoost models in L2 not only achieve higher median and peak MCC values but also demonstrate their strength in uncovering subtle and implicit patterns embedded in cardiovascular disease risk data. Convergence curves further highlight this advantage, with CatBoost displaying smooth and stable optimization dynamics across multiple metaheuristic strategies. The benefit is most pronounced when optimized with CDGPSO, where CatBoost records the highest MCC scores and markedly improves discriminative power for this challenging classification problem.

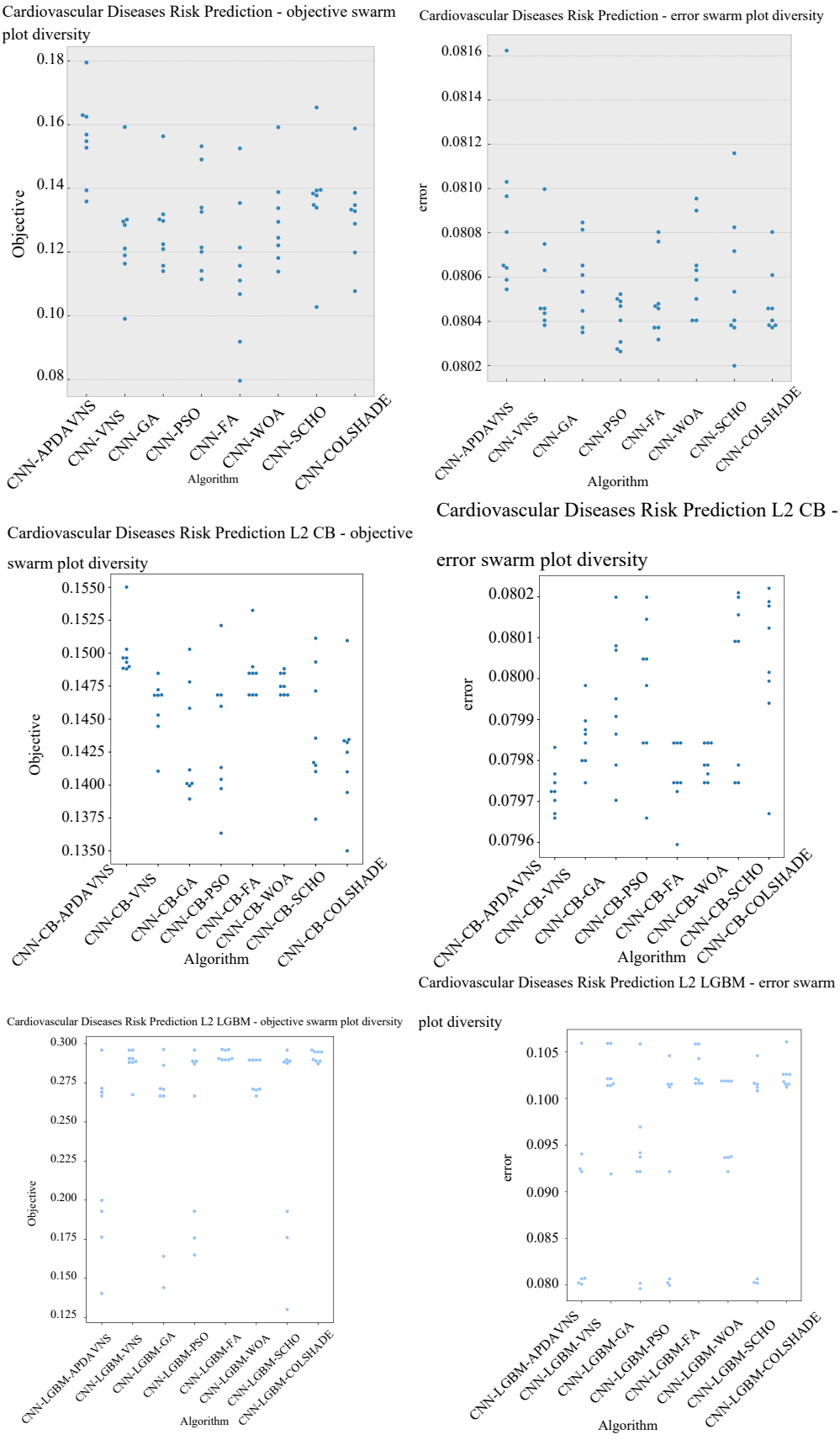


Figure 7. Swarm plots of both MCC and error for L1 CNN (**top**), L2 CatBoost (**middle**) and L2 LightGBM (**bottom**) simulations. Colors are used solely for visualization purposes, algorithms are identified by the x-axis labels.

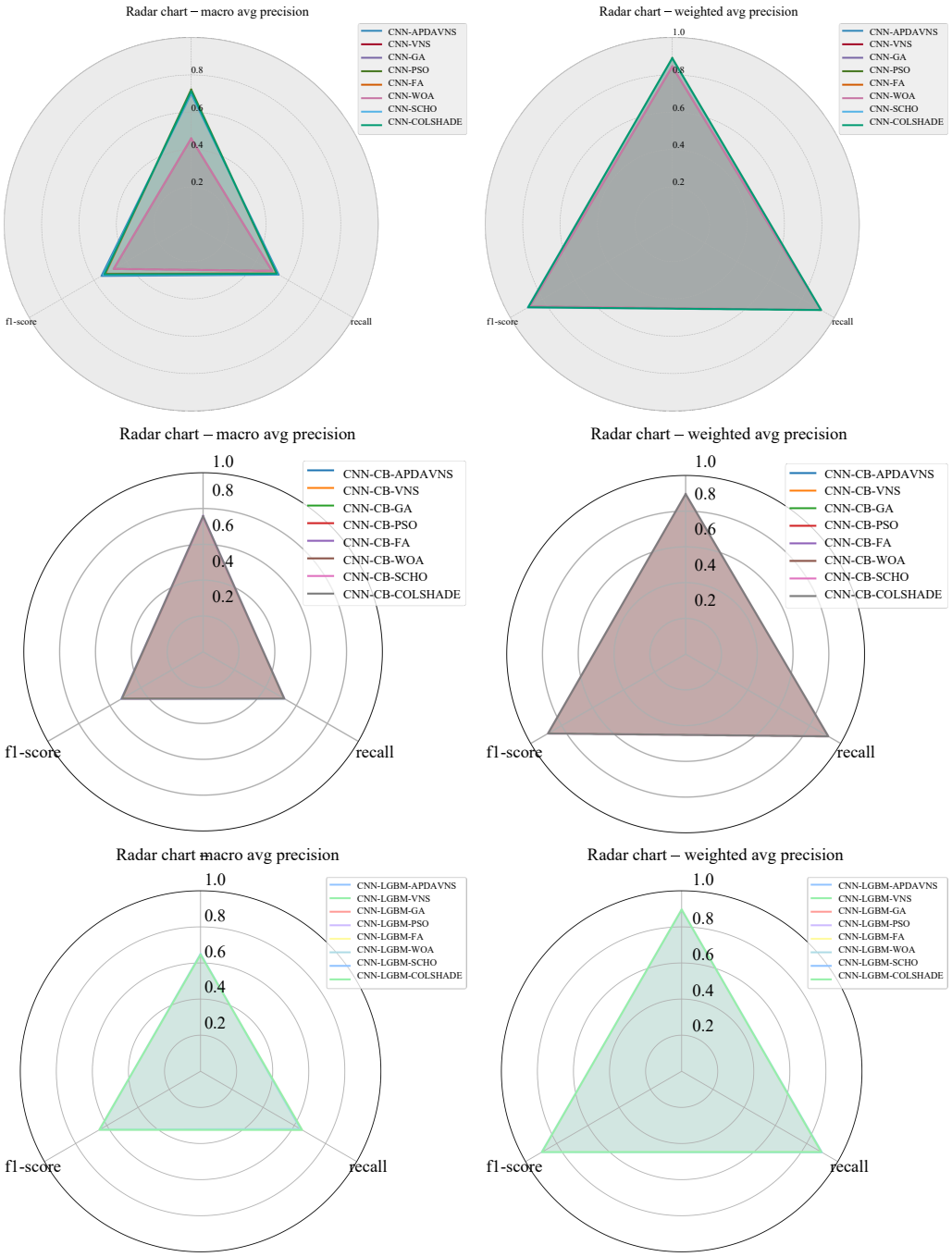


Figure 8. Radar diagrams of macro and weighted averages for L1 CNN (**top**), L2 CatBoost (**middle**) and L2 LightGBM (**bottom**) simulations. Colors indicate different algorithms, as specified in the legend.

A similar pattern emerges in the analysis of the indicator function (error rate), where lower values signify better performance. CatBoost at L2 consistently delivers some of the lowest error rates across all examined scenarios. Its box plots frequently show reduced medians and tighter variability, underscoring robustness in handling class imbalance and reducing misclassification. Although LightGBM at L2 provides competitive results and remains a practical, lightweight option, CatBoost distinguishes itself by offering the most dependable and generalizable performance across a range of metaheuristic optimizers. Collectively, these results affirm CatBoost as the preferred choice for the second layer in

CVD risk classification pipelines, combining high predictive accuracy with reduced error rates and improved stability under real-world clinical variability.

6. Results Validation, Comparisons with Other SOTA Approaches and Explainability

6.1. Statistical Analysis

Although single-run comparisons of optimization algorithms may offer some preliminary insights, they are generally unreliable given the stochastic nature of metaheuristics, where outcomes can vary widely between executions. To mitigate this variability and capture representative performance, the study executed 30 independent runs for each algorithm, each initialized with a different random seed. This procedure yields a distribution of results that reflects the algorithms' typical behavior rather than isolated outcomes. Such a multi-run evaluation not only facilitates statistically grounded comparisons but also enables rigorous hypothesis testing and clearer identification of performance patterns. Additionally, this technique adheres to recognized best practices in benchmarking metaheuristic methods [67], thereby enhancing both the reliability and reproducibility of the reported findings.

Statistical procedures for assessing whether group performance differences are meaningful are typically divided into two main categories: parametric and non-parametric. Parametric tests are contingent on several assumptions, including independence of observations, normally distributed data, and equality of variances across groups (homoscedasticity) [67]. In this study, independence was ensured by initializing each optimization run with a distinct random seed, thereby eliminating cross-run dependencies. To evaluate homoscedasticity, Levene's test [68] was applied, producing a p-value of 0.62 for all algorithms, which indicated no significant variance differences between groups. Subsequently, the assumption of normality was examined using the Shapiro–Wilk test [69], with the corresponding p-values listed in Table 16. Since all observed values were below the conventional 0.05 threshold, the null hypothesis of normality was rejected. This outcome confirmed that the normality condition required for parametric testing was not met, necessitating reliance on non-parametric alternatives.

Table 16. Shapiro–Wilk p-values for each metaheuristic and conducted experiment.

Model	APDAVNS	VNS	GA	PSO	FA	WOA	SCHO	COLSHADE
L1 CNN	0.026	0.023	0.031	0.027	0.033	0.032	0.029	0.036
L2 CatBoost	0.024	0.021	0.033	0.035	0.026	0.031	0.028	0.022
L2 LightGBM	0.019	0.021	0.023	0.016	0.014	0.011	0.022	0.018

The kernel density estimation (KDE) curves delineated within Figure 9 visually reinforce these findings, clearly illustrating significant deviations from normality. Consequently, all subsequent statistical analyses in this study were conducted using non-parametric methods, providing a more robust evaluation framework that does not depend on stringent distributional assumptions.

Given the rejection of the normality assumption, the analysis proceeded with a non-parametric approach. In particular, the Wilcoxon signed-rank test [70] was employed to carry out pairwise comparisons between the proposed APDAVNS algorithm and each of the contending metaheuristics. The resulting p-values, summarized in Table 17, were all below the conventional significance threshold $\alpha = 0.05$, confirming that APDAVNS outperformed every alternative method by a statistically significant margin.

These results demonstrate that the superiority of APDAVNS cannot be explained by random variability or sampling noise. Instead, they underscore consistent, substantive gains across all three experimental setups, consequently reinforcing both the robustness and the effectiveness of the proposed approach.

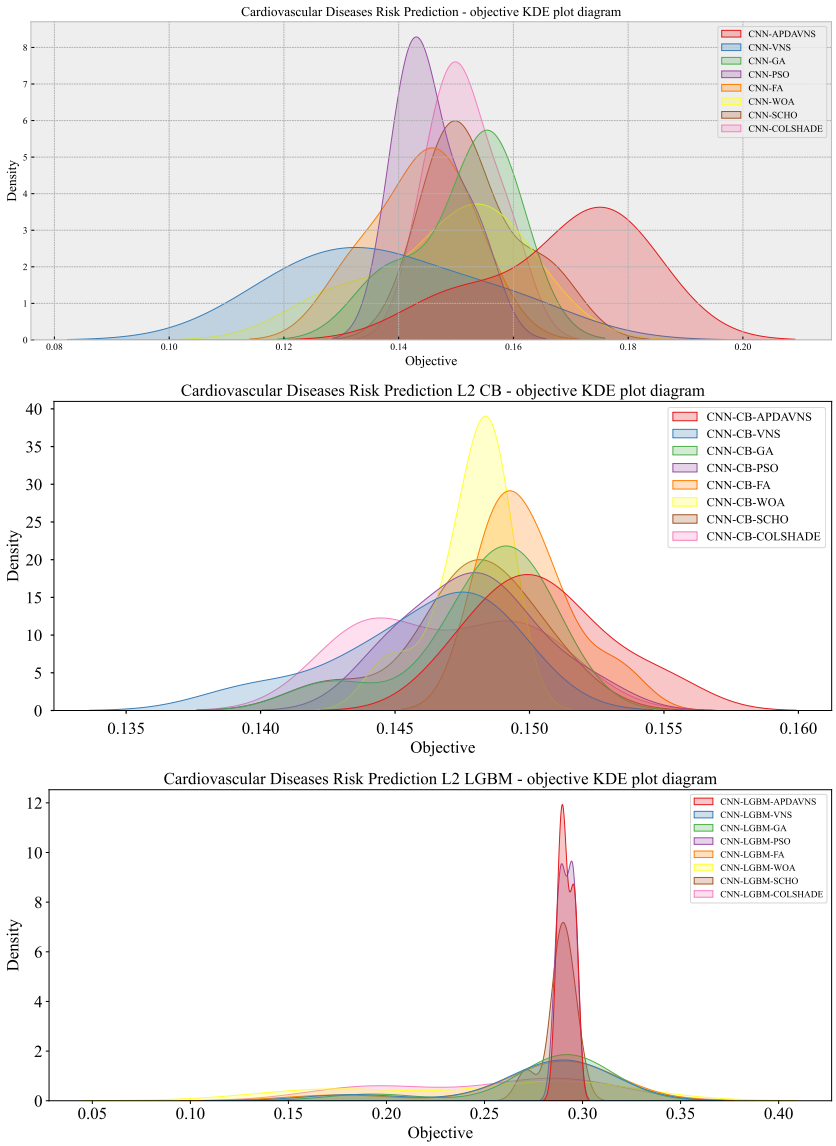


Figure 9. Fitness function KDE plots for L1 CNN (**top**) and L2 CatBoost (**middle**) and LightGBM (**bottom**) classifiers experiments.

Table 17. Wilcoxon signed-rank test evaluation (APDAVNS against other contenders) for all three conducted experiments.

APDAVNS vs. Others	VNS	GA	PSO	FA	WOA	SCHO	COLSHADE
L1 CNN	0.019	0.033	0.029	0.026	0.031	0.037	0.035
L2 CatBoost	0.025	0.037	0.034	0.039	0.031	0.033	0.027
L2 LightGBM	0.031	0.035	0.042	0.033	0.018	0.038	0.024

6.2. Comparisons with Baseline Models

The performance of the best tuned L2 models is further assessed against a competitive set of baseline models. This collection included well-known classifiers like multi-layer perceptron (MLP), decision tree, random forest, KNN along with ensemble and boosting

models like XGBoost, AdaBoost, baseline CatBoost and baseline LightGBM, and deep models like baseline CNN. All baseline models were configured with default hyperparameter settings. Baseline CNN was configured with a 1D convolutional layer with 32 neurons (filters), kernel size 3, activation function ReLU, the flatten layer (standard), followed by a dense layer with 128 neurons, and finally a dense layer with 2 neurons, since the problem is a binary classification task.

Table 18 presents a detailed comparison of the best CatBoost and LightGBM classifiers used in the second layer against the abovementioned baseline models. Among these, CNN-CB-APDAVNS produced the highest overall accuracy, attaining 0.92033. More importantly, the CNN-CB-APDAVNS method proved significantly more effective in terms of precision, recall (sensitivity), and F1-score, especially when evaluating the minority class associated with CVD risk. This highlights a crucial point again, as accuracy alone is insufficient for judging model quality in this context. Another noteworthy finding is that baseline models attained high accuracy in general; however, their capabilities of handling minority classes are extremely limited.

Table 18. Comprehensive assessment of the best generated models against baseline models.

Approach	Metric	No Risk	Risk	Accuracy	Macro Ave	Weight Ave
CNN-CB-APDAVNS	precision	0.922702	0.583975	0.920330	0.753339	0.895317
	recall	0.996830	0.050594	0.920330	0.523712	0.920330
	f1-score	0.958335	0.093120	0.920330	0.525728	0.888385
CNN-LGBM-APDAVNS	precision	0.943424	0.348934	0.894072	0.646179	0.895361
	recall	0.941197	0.358297	0.894072	0.649747	0.894072
	f1-score	0.942309	0.353553	0.894072	0.647931	0.894710
AdaBoost	precision	0.924457	0.481023	0.918657	0.702740	0.888607
	recall	0.992614	0.077827	0.918657	0.535221	0.918657
	f1-score	0.957324	0.133977	0.918657	0.545651	0.890759
CatBoost	precision	0.922594	0.468137	0.918592	0.695366	0.885853
	recall	0.994904	0.050995	0.918592	0.522949	0.918592
	f1-score	0.957386	0.091971	0.918592	0.524678	0.887420
CNN	precision	0.920401	0.607306	0.919661	0.763853	0.895088
	recall	0.998990	0.017755	0.919661	0.508372	0.919661
	f1-score	0.958087	0.034501	0.919661	0.496294	0.883418
Decision Tree	precision	0.930951	0.200120	0.865169	0.565535	0.871866
	recall	0.921671	0.222801	0.865169	0.572236	0.865169
	f1-score	0.926288	0.210852	0.865169	0.568570	0.868447
KNN	precision	0.926492	0.320961	0.908534	0.623727	0.877537
	recall	0.978090	0.117741	0.908534	0.547916	0.908534
	f1-score	0.951592	0.172282	0.908534	0.561937	0.888588
LightGBM	precision	0.921526	0.542194	0.919585	0.731860	0.890858
	recall	0.997452	0.034308	0.919585	0.515880	0.919585
	f1-score	0.957987	0.064532	0.919585	0.511260	0.885754
MLP	precision	0.922232	0.523438	0.919477	0.722835	0.889991
	recall	0.996419	0.044720	0.919477	0.520570	0.919477
	f1-score	0.957891	0.082401	0.919477	0.520146	0.887111
Random Forest	precision	0.924173	0.405674	0.916283	0.664923	0.882254
	recall	0.990160	0.076358	0.916283	0.533259	0.916283
	f1-score	0.956029	0.128525	0.916283	0.542277	0.889128
XGBoost	precision	0.921776	0.520000	0.919391	0.720888	0.889294
	recall	0.996900	0.038179	0.919391	0.517540	0.919391
	f1-score	0.957867	0.071135	0.919391	0.514501	0.886178
	samples	85,166	7491			

6.3. Best Model Interpretation

SHapley Additive exPlanations (SHAP) is a model-agnostic interpretability framework that attributes an importance score to every feature contributing to a prediction. Its foundation lies in cooperative game theory, drawing specifically on the concept of Shapley values [71], which offer a principled way of distributing a total outcome among participants according to their individual contributions. By doing so, SHAP consolidates multiple interpretability techniques into a unified additive formulation.

At its core, SHAP evaluates how much each feature influences a prediction relative to a baseline expectation. The contribution of a feature is formalized as its Shapley value ϕ_i, defined as the weighted mean of its marginal impacts when added to every possible subset of the remaining features. This construction ensures a fair and mathematically consistent decomposition of model outputs into feature-level attributions. Its mathematical formulation is provided in Equation (14).

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \Big[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S) \Big]. \tag{14}$$

In this formulation, F denotes the complete feature set, while S refers to any subset that does not include feature i. The term $f_S(x_S)$ corresponds to the model output restricted to the features contained in S. The factorial components serve as combinatorial weights, balancing contributions across every possible ordering of features and thereby guaranteeing an equitable allocation of importance [71].

SHAP values are particularly appealing because they adhere to several key theoretical properties:

- Local accuracy—the sum of all feature contributions precisely equals the difference between the model output and the baseline expectation.
- Missingness—any feature absent from the model input is assigned zero contribution.
- Consistency—if a model is adjusted such that a feature's influence increases, its SHAP value will not decrease.

Together, these properties establish SHAP as a rigorous and principled framework for model interpretability. In practice, it has become a widely adopted tool for explaining predictions of complex models in high-stakes domains such as healthcare and finance, where transparency is essential. SHAP enables both local interpretability (explaining individual predictions) and global analysis (assessing feature importance across an entire dataset). However, its exact computation scales exponentially with the number of features, motivating the development of approximation methods and specialized variants such as TreeSHAP for tree-structured models [72].

The strengths of SHAP are found in its model-agnostic design, its grounding in fairness through Shapley values, and its additive nature, which makes interpretation more intuitive. Nonetheless, several limitations persist: the method can be computationally intensive, certain implementations rely on independence assumptions that may not hold in practice, and approximation strategies can introduce variability. Moreover, meaningful interpretation of SHAP outputs still depends on substantial domain expertise, without which the attributions risk being misunderstood.

In addition to the general limitations of SHAP, the present analysis exhibits instance-level feature attributions that may contradict established clinical evidence. Most notably, diabetes mellitus appears with a negative SHAP contribution in certain instances, despite being a well-recognized independent risk factor for CVD. This behavior should not be interpreted as a protective or clinically meaningful effect. Rather, it reflects conditional associations learned by the model within the constraints of the dataset and the reduced

feature representation used for modeling. Such counterintuitive attributions may arise from dataset composition, feature interactions, unobserved confounders, or diabetes acting as a proxy for correlated variables such as healthcare utilization or age. Consequently, SHAP results in this study are reported to explain model behavior for specific instances, not to validate causal or population-level clinical relationships.

In this study, a hybrid SHAP procedure was employed to explain model behavior. The workflow begins with the original dataset, followed by feature extraction using the best-performing CNN identified in the L1 stage. During this process, the CNN is operated in inference mode, with dropout layers disabled, and feature embeddings are extracted at the output of the dropout layer to ensure deterministic representations. These embeddings are then provided as input to the top-performing CatBoost classifier obtained in the L2 stage.

Although CatBoost is trained on CNN-derived embeddings, SHAP analysis is performed on the frozen end-to-end CNN-CatBoost pipeline, allowing prediction contributions to be propagated back to the original input features. As a result, SHAP values are expressed in terms of the original feature names rather than latent embedding dimensions. SHAP values are computed to interpret the classifier's predictions, and the resulting SHAP visualizations for both outcome classes (class 0 - no CVD risk and class 1 - CVD risk) are presented in Figure 10.

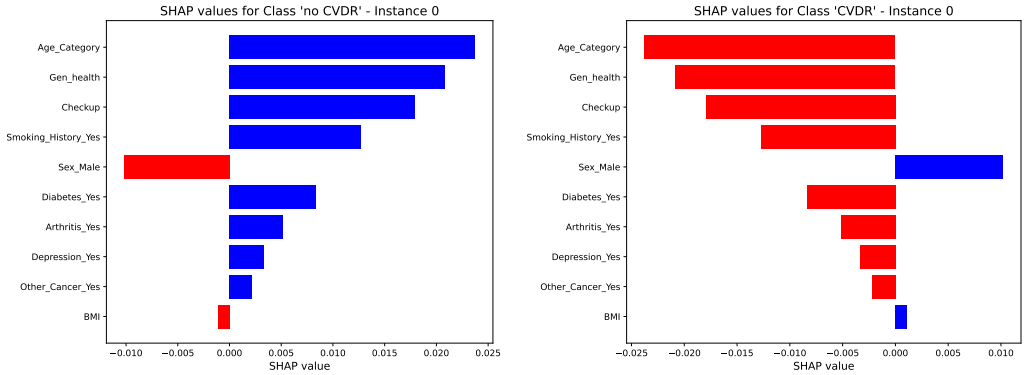


Figure 10. SHAP-based feature attribution for the frozen CNN-CatBoost pipeline, expressed with respect to the original input features, for class 0 (no CVD risk) and class 1 (CVD risk).

6.3.1. SHAP Interpretation for Class 0 (No CVD Risk)

The following interpretation corresponds to an instance-level SHAP explanation and describes how the trained model arrived at a no CVD risk prediction for a specific individual. The reported contributions reflect model behavior conditioned on the training data and should not be interpreted as causal or population-level risk relationships.

For the no CVD risk classification (class 0), the feature with the largest absolute SHAP contribution was the patient's age, indicating that, for this individual, age played the dominant role in shifting the model's prediction toward the no-risk class. This attribution reflects how the learned model associates the given age category with lower predicted risk when conditioned on the remaining feature values. Although increasing age is a well-established cardiovascular risk factor at the population level, associated with vascular stiffening, endothelial dysfunction, and cumulative exposure to risk factors, as demonstrated in longitudinal studies such as Framingham and CARDIA [3,5,73], its SHAP contribution here should be understood strictly as a model-specific effect for this instance.

Self-reported general health also exhibited a substantial negative SHAP value, contributing to the no CVD classification. This suggests that, within the model, better perceived health status is associated with lower predicted risk for this individual. Prior studies have shown that self-rated health correlates with long-term cardiovascular outcomes and mor-

tality [74,75]; however, in this context, the SHAP attribution reflects the model's learned associations rather than a direct protective mechanism. A similar effect was observed for regular medical check-ups, which modestly shifted the prediction toward the no-risk class. Preventive health behaviors, including routine primary care visits, are often associated with improved detection and management of modifiable risk factors [76], though their direct impact on cardiovascular outcomes remains debated.

Several traditional cardiovascular risk factors, including smoking history and diabetes mellitus, showed small negative SHAP contributions for this instance, indicating that the model associated their presence with a slight decrease in predicted CVD risk. These directions are counterintuitive and inconsistent with extensive evidence linking tobacco exposure and diabetes to atherosclerosis, myocardial infarction, and stroke [77–80]. Such findings are unlikely to represent true protective effects and instead likely reflect dataset-specific correlations, conditioning effects, or interactions with other variables.

Additional comorbidities such as arthritis, depression, and a history of cancer also contributed modestly toward the no-risk classification. Clinically, these conditions have been associated with increased cardiovascular risk through pathways involving chronic inflammation, behavioral factors, or treatment-related cardiotoxicity [81–84]. Their negative SHAP contributions here therefore highlight model-specific associations within the dataset rather than clinically meaningful protective relationships.

Male sex exhibited a positive SHAP value, shifting the prediction modestly toward higher cardiovascular risk, consistent with epidemiological patterns showing earlier onset and higher incidence of coronary heart disease in men compared to women [85,86]. Body mass index (BMI), by contrast, exerted only a small influence in this case, suggesting that its contribution was either within a low-risk range or outweighed by other features with stronger influence on the model's decision [87].

6.3.2. SHAP Interpretation for Class 1 (CVD Risk)

The following interpretation reflects an instance-level SHAP explanation for a prediction of CVD risk (class 1). The contributions describe how individual features influenced the model's output for this specific case and do not imply generalizable or causal relationships.

For the CVD risk classification (class 1), only a limited number of features contributed positively toward the predicted risk. In particular, male sex and body mass index (BMI) were the primary variables pushing the model's prediction toward the CVD risk class. These features exhibited the largest positive SHAP values, indicating that, for this individual, they represented the most influential risk-enhancing signals within the learned model. This pattern is broadly consistent with clinical evidence linking male sex and elevated BMI to increased cardiovascular risk through mechanisms such as hypertension, insulin resistance, and dyslipidaemia [85–87].

In contrast, most remaining features contributed negatively, shifting the prediction away from the CVD risk class. This indicates that, for this individual, the model detected several characteristics that it associated with lower predicted risk, partially offsetting the influence of sex and BMI. Such competing contributions illustrate that the final prediction emerged from a balance of multiple feature effects rather than uniformly elevated risk signals.

Notably, smoking history and diabetes mellitus again showed negative SHAP contributions in this instance. As in the no-risk case, these directions are inconsistent with established clinical knowledge [77–80] and should not be interpreted as biologically meaningful. Instead, they likely reflect dataset structure, feature interactions, or correlations with other health and demographic variables present in the training data.

Overall, the SHAP explanation for the CVD risk case highlights that the model's prediction was driven by a small subset of risk-enhancing features, while many other variables exerted mitigating effects. This underscores both the complexity of the model's decision process and the importance of interpreting SHAP values as model-specific explanations, particularly when applied to cross-sectional, self-reported health datasets.

7. Conclusions

The integration of robust CVD risk classification methods into healthcare decision-making frameworks carries profound policy implications for health systems. By enabling accurate and timely risk stratification, these models can inform policies related to prevention, triage, and resource allocation across the continuum of care. For example, predictive insights from ML models can guide the prioritization of clinical monitoring, specialist consultations, and preventive interventions, particularly during times of resource strain like public health crises or surges in patient volume. Policymakers can also employ these tools to design adaptive strategies that allocate resources dynamically in response to shifting population risk profiles, thereby improving both efficiency and patient outcomes.

Beyond immediate operational impacts, the adoption of CVD risk classification models holds long-term significance for health system planning and policy development. Risk prediction can serve as the foundation for standardized, evidence-based care pathways, helping reduce variability across institutions and ensuring equitable access to timely interventions for high-risk patients. Moreover, the population-level data produced by these models can inform national prevention strategies, guide public health campaigns, and shape reimbursement structures tied to outcomes and preventive effectiveness. To unlock their full potential, however, policy frameworks must also address the ethical, legal, and technical requirements for integrating AI-driven tools into clinical practice, ranging from data security and algorithm transparency to clinician training and patient trust.

Accurate methods for CVD risk prediction are essential for anticipating patient trajectories and guiding timely interventions. In high-stakes clinical contexts, the ability to reliably stratify individuals by their risk levels enables more effective prioritization of care, targeted allocation of resources, and improved health outcomes. Advanced classification models can capture subtle, nonlinear patterns within complex clinical datasets, providing insights beyond what traditional approaches can achieve. To address these demands, this study proposed a two-tier hybrid framework that combines CNN with gradient-boosting classifiers CatBoost and LightGBM. Model performance and parameter optimization were further enhanced through the use of metaheuristic algorithms, including a novel optimizer derived from VNS and adapted specifically for this task. Applied to a real-world dataset of patient records, the framework was evaluated in a binary CVD risk prediction setting. The most effective configuration in the simulations was the CNN feature extraction followed by CatBoost refined with the proposed APDAVNS algorithm, which achieved an impressive classification accuracy of slightly over 92%, while maintaining high precision for the majority class, with improved, but still limited, recognition of the minority class.

A comprehensive statistical analysis confirmed the superiority of the proposed method compared with both standard VNS and other optimization strategies. To enhance interpretability, SHAP was employed on the best-performing model. Feature vectors extracted from the CNN after dropout were supplied to CatBoost, and SHAP values were computed to highlight the contribution of individual predictors, consequently offering meaningful explanations of the model's decision logic.

This framework demonstrated several advantages. The proposed APDAVNS optimizer consistently surpassed cutting-edge alternatives, and the two-level architecture delivered better results than baseline CNNs without incurring excessive computational

cost. From an application standpoint, this kind of approach holds considerable promise for clinical adoption in cardiovascular care. It has the potential to enable earlier identification of individuals at elevated risk for adverse outcomes, including disease progression, complications, or mortality, and to support more effective triage and intervention strategies. In doing so, it could meaningfully enhance predictive accuracy and ultimately improve patient management.

Despite these gains, the results also reveal important limitations. Although high precision is achieved for the majority class, the recall for the minority class (CVD risk) remains comparatively low, reflecting the inherent challenges posed by severe class imbalance and the use of cross-sectional, self-reported data. In addition, the interpretability analysis identified instance-level feature attributions that do not always align with established clinical evidence, most notably for diabetes mellitus. These counterintuitive attributions reflect model and dataset-specific associations learned under a reduced feature representation and should not be interpreted as clinically meaningful or causal. Accordingly, SHAP results in this study are intended to explain model behavior rather than to validate known CVD risk factors. These findings indicate that, although the proposed framework enhances classification performance, it should not be interpreted as a fully ready and reliable clinical decision-support system. Instead, it serves as a proof-of-concept demonstrating the potential of staged metaheuristic optimization for risk modeling. Future work will focus on improving minority-class sensitivity, incorporating richer clinical variables, and validating the approach on longitudinal datasets to enhance clinical applicability.

Additionally, certain limitations of the study must also be recognized. The comparative analysis was restricted to a relatively narrow set of optimization algorithms and constrained by small population sizes and limited iteration counts. Future research will aim to expand the scope of optimizers, leverage larger datasets, and explore the scalability of the framework under more demanding experimental conditions. Moreover, the custom APDAVNS introduced here could be extended to a wider variety of ML applications, providing a versatile tool for hyperparameter optimization across domains. Applying this methodology to real-time streaming data represents another promising avenue for advancing precision medicine in cardiovascular disease.

Author Contributions: P.D., T.Z. and M.Z.—writing original draft, visualization; S.J.V., S.A., M.M. and V.S.—writing—review and editing, validation; M.A.-S. and N.B.—simulations, statistical analysis, funding acquisition. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The study employs an openly available dataset, accessible via Kaggle at https://www.kaggle.com/datasets/alphiree/cardiovascular-diseases-risk-prediction-dataset (accessed on 15 November 2025).

Acknowledgments: This research was supported by the Science Fund of the Republic of Serbia, Grant No. 7373, Characterizing crises-caused air pollution alternations using an artificial intelligence-based framework—crAIRsis and Grant No. 7502, Intelligent Multi-Agent Control and Optimization applied to Green Buildings and Environmental Monitoring Drone Swarms—ECOSwarm.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Roth, G.A.; Mensah, G.A.; Johnson, C.O.; Addolorato, G.; Ammirati, E.; Baddour, L.M.; Barengo, N.C.; Beaton, A.Z.; Benjamin, E.J.; Benziger, C.P.; et al. Global burden of cardiovascular diseases and risk factors, 1990–2019: Update from the GBD 2019 study. *J. Am. Coll. Cardiol.* **2020**, *76*, 2982–3021. [CrossRef] [PubMed]
2. Di Cesare, M.; Perel, P.; Taylor, S.; Kabudula, C.; Bixby, H.; Gaziano, T.A.; McGhie, D.V.; Mwangi, J.; Pervan, B.; Narula, J.; et al. The heart of the world. *Glob. Heart* **2024**, *19*, 11. [CrossRef]
3. D'Agostino, R.B., Sr.; Vasan, R.S.; Pencina, M.J.; Wolf, P.A.; Cobain, M.; Massaro, J.M.; Kannel, W.B. General cardiovascular risk profile for use in primary care: The Framingham Heart Study. *Circulation* **2008**, *117*, 743–753. [CrossRef]
4. Dawber, T.R.; Meadors, G.F.; Moore, F.E., Jr. Epidemiological approaches to heart disease: The Framingham Study. *Am. J. Public Health* **1951**, *41*, 279–286. [CrossRef]
5. Mahmood, S.S.; Levy, D.; Vasan, R.S.; Wang, T.J. The Framingham Heart Study and the epidemiology of cardiovascular disease: A historical perspective. *Lancet* **2014**, *383*, 999–1008. [CrossRef]
6. Conroy, R.M.; Pyörälä, K.; Fitzgerald, A.P.; Sans, S.; Menotti, A.; De Backer, G.; De Bacquer, D.; Ducimetière, P.; Jousilahti, P.; Keil, U.; et al. Estimation of ten-year risk of fatal cardiovascular disease in Europe: The SCORE project. *Eur. Heart J.* **2003**, *24*, 987–1003. [CrossRef]
7. Goff, D.C., Jr.; Lloyd-Jones, D.M.; Bennett, G.; Coady, S.; D'Agostino, R.B.; Gibbons, R.; Greenland, P.; Lackland, D.T.; Levy, D.; O'Donnell, C.J.; et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: A report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *J. Am. Coll. Cardiol.* **2014**, *63*, 2935–2959. [CrossRef]
8. Yadlowsky, S.; Hayward, R.A.; Sussman, J.B.; McClelland, R.L.; Min, Y.I.; Basu, S. Clinical implications of revised pooled cohort equations for estimating atherosclerotic cardiovascular disease risk. *Ann. Intern. Med.* **2018**, *169*, 20–29. [CrossRef]
9. Hippisley-Cox, J.; Coupland, C.; Brindle, P. Development and validation of QRISK3 risk prediction algorithms to estimate future risk of cardiovascular disease: Prospective cohort study. *BMJ* **2017**, *357*, j2099. [CrossRef]
10. DeFilippis, A.P.; Young, R.; Carrubba, C.J.; McEvoy, J.W.; Budoff, M.J.; Blumenthal, R.S.; Kronmal, R.A.; McClelland, R.L.; Nasir, K.; Blaha, M.J. An analysis of calibration and discrimination among multiple cardiovascular risk scores in a modern multiethnic cohort. *Ann. Intern. Med.* **2015**, *162*, 266–275. [CrossRef]
11. Nwanosike, E.M.; Conway, B.R.; Merchant, H.A.; Hasan, S.S. Potential applications and performance of machine learning techniques and algorithms in clinical practice: A systematic review. *Int. J. Med. Inform.* **2022**, *159*, 104679. [CrossRef]
12. Beam, A.L.; Kohane, I.S. Big data and machine learning in health care. *JAMA* **2018**, *319*, 1317–1318. [CrossRef]
13. Daghistani, T.A.; Elshawi, R.; Sakr, S.; Ahmed, A.M.; Al-Thwayee, A.; Al-Mallah, M.H. Predictors of in-hospital length of stay among cardiac patients: A machine learning approach. *Int. J. Cardiol.* **2019**, *288*, 140–147. [CrossRef]
14. Bollepalli, S.C.; Sahani, A.K.; Aslam, N.; Mohan, B.; Kulkarni, K.; Goyal, A.; Singh, B.; Singh, G.; Mittal, A.; Tandon, R.; et al. An optimized machine learning model accurately predicts in-hospital outcomes at admission to a cardiac unit. *Diagnostics* **2022**, *12*, 241. [CrossRef]
15. Krittanawong, C.; Johnson, K.W.; Rosenson, R.S.; Wang, Z.; Aydar, M.; Baber, U.; Min, J.K.; Tang, W.W.; Halperin, J.L.; Narayan, S.M. Deep learning for cardiovascular medicine: A practical primer. *Eur. Heart J.* **2019**, *40*, 2058–2073. [CrossRef] [PubMed]
16. Mpanya, D.; Celik, T.; Klug, E.; Ntsinjana, H. Predicting in-hospital all-cause mortality in heart failure using machine learning. *Front. Cardiovasc. Med.* **2023**, *9*, 1032524. [CrossRef]
17. Choi, M.H.; Kim, D.; Choi, E.J.; Jung, Y.J.; Choi, Y.J.; Cho, J.H.; Jeong, S.H. Mortality prediction of patients in intensive care units using machine learning algorithms based on electronic health records. *Sci. Rep.* **2022**, *12*, 7180. [CrossRef]
18. Sun, Y.; He, Z.; Ren, J.; Wu, Y. Prediction model of in-hospital mortality in intensive care unit patients with cardiac arrest: A retrospective analysis of MIMIC-IV database based on machine learning. *BMC Anesthesiol.* **2023**, *23*, 178. [CrossRef]
19. Ambale-Venkatesh, B.; Yang, X.; Wu, C.O.; Liu, K.; Hundley, W.G.; McClelland, R.; Gomes, A.S.; Folsom, A.R.; Shea, S.; Guallar, E.; et al. Cardiovascular event prediction by machine learning: The multi-ethnic study of atherosclerosis. *Circ. Res.* **2017**, *121*, 1092–1101. [CrossRef]
20. Wolpert, D.; Macready, W. No free lunch theorems for optimization. *IEEE Trans. Evol. Comput.* **1997**, *1*, 67–82. [CrossRef]
21. Villoth, J.P.; Zivkovic, M.; Zivkovic, T.; Abdel-salam, M.; Hammad, M.; Jovanovic, L.; Simic, V.; Bacanin, N. Two-tier deep and machine learning approach optimized by adaptive multi-population firefly algorithm for software defects prediction. *Neurocomputing* **2025**, *630*, 129695. [CrossRef]
22. Antonijevic, M.; Zivkovic, M.; Djuric Jovicic, M.; Nikolic, B.; Perisic, J.; Milovanovic, M.; Jovanovic, L.; Abdel-Salam, M.; Bacanin, N. Intrusion detection in metaverse environment internet of things systems by metaheuristics tuned two level framework. *Sci. Rep.* **2025**, *15*, 3555. [CrossRef] [PubMed]
23. Prokhorenkova, L.; Gusev, G.; Vorobev, A.; Dorogush, A.V.; Gulin, A. CatBoost: Unbiased boosting with categorical features. *Adv. Neural Inf. Process. Syst.* **2018**, *31*, 6639–6649.
24. Ke, G.; Meng, Q.; Finley, T.; Wang, T.; Chen, W.; Ma, W.; Ye, Q.; Liu, T.Y. Lightgbm: A highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* **2017**, *30*, 3149–3157.

25. Mladenović, N.; Hansen, P. Variable neighborhood search. *Comput. Oper. Res.* **1997**, *24*, 1097–1100. [CrossRef]
26. Turki, A.T.; Mussetti, A.; Scheidler, K.M.; Kehrmann, J.; Crivello, P.; Hosch, R.; Nensa, F.; Polizzi, S. Artificial intelligence in hematopoietic stem cell transplantation care and complication management. *Acta Haematol.* **2025**, *148*, 528–547. [CrossRef]
27. Bilal, D.A.; Alzahrani, A.; Almohammadi, K.; Saleem, M.; Farooq, M.S.; Sarwar, R. Explainable AI-driven intelligent system for precision forecasting in cardiovascular disease. *Front. Med.* **2025**, *12*, 1596335. [CrossRef]
28. Teshale, A.B.; Htun, H.L.; Vered, M.; Owen, A.J.; Freak-Poli, R. A systematic review of artificial intelligence models for Time-to-Event outcome applied in cardiovascular disease risk prediction. *J. Med. Syst.* **2024**, *48*, 68. [CrossRef]
29. Cai, Y.; Cai, Y.Q.; Tang, L.Y.; Wang, Y.H.; Gong, M.; Jing, T.C.; Li, H.J.; Li-Ling, J.; Hu, W.; Yin, Z.; et al. Artificial intelligence in the risk prediction models of cardiovascular disease and development of an independent validation screening tool: A systematic review. *BMC Med.* **2024**, *22*, 56. [CrossRef]
30. Liu, T.; Krentz, A.; Lu, L.; Curcin, V. Machine learning based prediction models for cardiovascular disease risk using electronic health records data: Systematic review and meta-analysis. *Eur. Heart J.-Digit. Health* **2025**, *6*, 7–22. [CrossRef]
31. Weng, S.F.; Reps, J.; Kai, J.; Garibaldi, J.M.; Qureshi, N. Can machine-learning improve cardiovascular risk prediction using routine clinical data? *PLoS ONE* **2017**, *12*, e0174944. [CrossRef]
32. Alaa, A.M.; Bolton, T.; Di Angelantonio, E.; Rudd, J.H.; Van der Schaar, M. Cardiovascular disease risk prediction using automated machine learning: A prospective study of 423,604 UK Biobank participants. *PLoS ONE* **2019**, *14*, e0213653. [CrossRef]
33. Teshale, A.B.; Htun, H.L.; Vered, M.; Owen, A.J.; Ryan, J.; Tonkin, A.; Freak-Poli, R. Artificial intelligence improves risk prediction in cardiovascular disease. *GeroScience* **2025**, *47*, 5497–5502. [CrossRef]
34. Wang, Y.; Aivalioti, E.; Stamatelopoulos, K.; Zervas, G.; Mortensen, M.B.; Zeller, M.; Liberale, L.; Di Vece, D.; Schweiger, V.; Camici, G.G.; et al. Machine learning in cardiovascular risk assessment: Towards a precision medicine approach. *Eur. J. Clin. Investig.* **2025**, *55*, e70017. [CrossRef] [PubMed]
35. Kasartzian, D.I.; Tsiampalis, T. Transforming cardiovascular risk prediction: A review of machine learning and artificial intelligence innovations. *Life* **2025**, *15*, 94. [CrossRef] [PubMed]
36. Huang, W.; Ying, T.W.; Chin, W.L.C.; Baskaran, L.; Marcus, O.E.H.; Yeo, K.K.; Kiong, N.S. Application of ensemble machine learning algorithms on lifestyle factors and wearables for cardiovascular risk prediction. *Sci. Rep.* **2022**, *12*, 1033. [CrossRef]
37. Delpino, F.; Costa, Â.; Farias, S.; Chiavegatto Filho, A.D.P.; Arcêncio, R.A.; Nunes, B. Machine learning for predicting chronic diseases: A systematic review. *Public Health* **2022**, *205*, 14–25. [CrossRef]
38. Damen, J.A.; Hooft, L.; Schuit, E.; Debray, T.P.; Collins, G.S.; Tzoulaki, I.; Lassale, C.M.; Siontis, G.C.; Chiocchia, V.; Roberts, C.; et al. Prediction models for cardiovascular disease risk in the general population: Systematic review. *BMJ* **2016**, *353*, i2416. [CrossRef]
39. Tsai, M.L.; Chen, K.F.; Chen, P.C. Harnessing electronic health records and artificial intelligence for enhanced cardiovascular risk prediction: A comprehensive review. *J. Am. Heart Assoc.* **2025**, *14*, e036946. [CrossRef]
40. Gu, J.; Wang, Z.; Kuen, J.; Ma, L.; Shahroudy, A.; Shuai, B.; Liu, T.; Wang, X.; Wang, G.; Cai, J.; et al. Recent advances in convolutional neural networks. *Pattern Recognit.* **2018**, *77*, 354–377. [CrossRef]
41. Bezdan, T.; Milosevic, S.; Zivkovic, M.; Bacanin, N.; Strumberger, I.; et al. Optimizing convolutional neural network by hybridized elephant herding optimization algorithm for magnetic resonance image classification of glioma brain tumor grade. In Proceedings of the 2021 Zooming Innovation in Consumer Technologies Conference (ZINC), Novi Sad, Serbia, 26–27 May 2021; IEEE: New York, NY, USA, 2021; pp. 171–176.
42. Islam, M.A.; Rabbi, F.; Hossain, N.U.I. Performance evaluation of NLP and CNN models for disaster detection using social media data. *Soc. Netw. Anal. Min.* **2024**, *14*, 213. [CrossRef]
43. Tasic, A.; Jovanovic, L.; Bacanin, N.; Zivkovic, M.; Simic, V.; Popovic, M.; Antonijevic, M. Towards sustainable societies: Convolutional neural networks optimized by modified crayfish optimization algorithm aided by AdaBoost and XGBoost for waste classification tasks. *Appl. Soft Comput.* **2025**, *175*, 113086. [CrossRef]
44. Swetha, T.; Roopa, R.; Sajitha, T.; Vidhyashree, B.; Sravani, J.; Praveen, B. Forecasting online shoppers purchase intentions with cat boost classifier. In Proceedings of the 2024 International Conference on Distributed Computing and Optimization Techniques (ICDCOT), Bengaluru, India, 15–16 March 2024; IEEE: New York, NY, USA, 2024; pp. 1–6.
45. Geeitha, S.; Ravishankar, K.; Cho, J.; Easwaramoorthy, S.V. Integrating cat boost algorithm with triangulating feature importance to predict survival outcome in recurrent cervical cancer. *Sci. Rep.* **2024**, *14*, 19828. [CrossRef]
46. Lokker, C.; Abdelkader, W.; Bagheri, E.; Parrish, R.; Cotoi, C.; Navarro, T.; Germini, F.; Linkins, L.A.; Haynes, R.B.; Chu, L.; et al. Boosting efficiency in a clinical literature surveillance system with LightGBM. *PLoS Digit. Health* **2024**, *3*, e0000299. [CrossRef]
47. Zhou, S.; Zhang, D.; Wang, M.; Liu, Z.; Gan, W.; Zhao, Z.; Xue, S.; Müller, B.; Zhou, M.; Ni, X.; et al. Risk-driven composition decoupling analysis for urban flooding prediction in high-density urban areas using Bayesian-Optimized LightGBM. *J. Clean. Prod.* **2024**, *457*, 142286. [CrossRef]
48. Huang, C.; Cai, Y.; Cao, J.; Deng, Y. Stock complex networks based on the GA-LightGBM model: The prediction of firm performance. *Inf. Sci.* **2025**, *700*, 121824. [CrossRef]

49. Mirjalili, S. Genetic algorithm. In *Evolutionary Algorithms and Neural Networks: Theory and Applications*; Springer: Cham, Switzerland, 2019; pp. 43–55.
50. Kennedy, J.; Eberhart, R. Particle swarm optimization. In Proceedings of the ICNN'95—International Conference on Neural Networks, Perth, WA, Australia, 27 November–1 December 1995; Volume 4, pp. 1942–1948.
51. Villoth, S.J.; Villoth, J.P.; Jovanovic, L.; Mani, J.; Zivkovic, M.; Zivkovic, T.; Bacanin, N.; Antonijevic, M.; Milovanovic, M. Optimizing Error Detection in Generated Code Using Metaheuristic Optimized Natural Language Processing. In *Soft Computing and Its Engineering Applications*; Springer: Cham, Switzerland, 2025; pp. 239–253.
52. Jovanovic, L.; Bacanin, N.; Zivkovic, M.; Antonijevic, M.; Petrovic, A.; Zivkovic, T. Anomaly detection in ECG using recurrent networks optimized by modified metaheuristic algorithm. In *2023 31st Telecommunications Forum (TELFOR)*; IEEE: New York, NY, USA, 2023; pp. 1–4.
53. Zivkovic, M.; Bacanin, N.; Zivkovic, T.; Jovanovic, L.; Kaljevic, J.; Antonijevic, M. Parkinson's Detection From Gait Time Series Classification Using LSTM Tuned by Modified RSA Algorithm. In Proceedings of the International Conference on Communication and Computational Technologies, Wuhan, China, 24–25 September 2023; Springer: Singapore, 2023; pp. 119–134.
54. Zivkovic, M.; Antonijevic, M.; Jovanovic, L.; Krasic, M.; Bacanin, N.; Zivkovic, T.; Milovanovic, M. Ocular Disease Diagnosis Using CNNs Optimized by Modified Variable Neighborhood Search Algorithm. In *Proceedings of International Joint Conference on Advances in Computational Intelligence*; Springer: Singapore, 2024; pp. 99–112.
55. Bacanin, N.; Jovanovic, L.; Djordjevic, M.; Petrovic, A.; Zivkovic, T.; Zivkovic, M.; Antonijevic, M. Crop yield forecasting based on echo state network tuned by crayfish optimization algorithm. In *2024 IEEE International Conference on Contemporary Computing and Communications (InC4)*; IEEE: New York, NY, USA, 2024; Volume 1, pp. 1–6.
56. Lakicevic, B.; Spalevic, Z.; Volas, I.; Jovanovic, L.; Zivkovic, M.; Zivkovic, T.; Bacanin, N. Artificial Neural Networks with Soft Attention: Natural Language Processing for Phishing Email Detection Optimized with Modified Metaheuristics. In *Advanced Network Technologies and Intelligent Computing*; Springer: Cham, Switzerland, 2024; pp. 421–438.
57. Antonijevic, M.; Jovanovic, L.; Bacanin, N.; Zivkovic, M.; Kaljevic, J.; Zivkovic, T. Using BERT with modified metaheuristic optimized XGBoost for phishing email identification. In *Proceedings of 4th International Conference on Artificial Intelligence and Smart Energy*; Springer: Cham, Switzerland, 2024; pp. 358–370.
58. Dobrojevic, M.; Jovanovic, L.; Babic, L.; Cajic, M.; Zivkovic, T.; Zivkovic, M.; Muthusamy, S.; Antonijevic, M.; Bacanin, N. Cyberbullying Sexism Harassment Identification by Metaheurustics-Tuned eXtreme Gradient Boosting. *Comput. Mater. Contin.* **2024**, *80*, 4997–5027. [CrossRef]
59. Mladenovic, D.; Antonijevic, M.; Jovanovic, L.; Simic, V.; Zivkovic, M.; Bacanin, N.; Zivkovic, T.; Perisic, J. Sentiment classification for insider threat identification using metaheuristic optimized machine learning classifiers. *Sci. Rep.* **2024**, *14*, 25731. [CrossRef]
60. Cheng, S.; Shi, Y. Diversity control in particle swarm optimization. In Proceedings of the 2011 IEEE Symposium on Swarm Intelligence, Paris, France, 11–15 April 2011; pp. 1–9. [CrossRef]
61. Lupague, R.; Mabborang, R.C.; Bansil, A.G.; Lupague, M.M.; et al. Integrated machine learning model for comprehensive heart disease risk assessment based on multi-dimensional health factors. *Eur. J. Comput. Sci. Inf. Technol.* **2023**, *11*, 44–58. [CrossRef]
62. Matthews, B.W. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. *Biochim. Biophys. Acta (BBA)-Protein Struct.* **1975**, *405*, 442–451. [CrossRef]
63. Yang, X.S.; He, X. Firefly algorithm: Recent advances and applications. *Int. J. Swarm Intell.* **2013**, *1*, 36–50. [CrossRef]
64. Mirjalili, S.; Lewis, A. The Whale Optimization Algorithm. *Adv. Eng. Softw.* **2016**, *95*, 51–67. [CrossRef]
65. Bai, J.; Li, Y.; Zheng, M.; Khatir, S.; Benaissa, B.; Abualigah, L.; Abdel Wahab, M. A Sinh Cosh optimizer. *Knowl.-Based Syst.* **2023**, *282*, 111081. [CrossRef]
66. Gurrola-Ramos, J.; Hernàndez-Aguirre, A.; Dalmau-Cedeño, O. COLSHADE for Real-World Single-Objective Constrained optimization Problems. In Proceedings of the 2020 IEEE Congress on Evolutionary Computation (CEC), Glasgow, UK, 19–24 July 2020; pp. 1–8.
67. LaTorre, A.; Molina, D.; Osaba, E.; Poyatos, J.; Del Ser, J.; Herrera, F. A prescription of methodological guidelines for comparing bio-inspired optimization algorithms. *Swarm Evol. Comput.* **2021**, *67*, 100973. [CrossRef]
68. Schultz, B.B. Levene's Test for Relative Variation. *Syst. Biol.* **1985**, *34*, 449–456. [CrossRef]
69. Shapiro, S.S.; Francia, R.S. An Approximate Analysis of Variance Test for Normality. *J. Am. Stat. Assoc.* **1972**, *67*, 215–216. [CrossRef]
70. Woolson, R.F. Wilcoxon Signed-Rank Test. In *Encyclopedia of Biostatistics*; John Wiley & Sons, Inc.: Hoboken, NJ, USA, 2005. [CrossRef]
71. Lundberg, S.M.; Lee, S.I. A unified approach to interpreting model predictions. In Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17), Long Beach, CA, USA, 4–9 December 2017; pp. 4768–4777.
72. Lundberg, S.M.; Erion, G.; Chen, H.; DeGrave, A.; Prutkin, J.M.; Nair, B.; Katz, R.; Himmelfarb, J.; Bansal, N.; Lee, S.I. From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.* **2020**, *2*, 56–67. [CrossRef]

73. Lloyd-Jones, D.M.; Lewis, C.E.; Schreiner, P.J.; Shikany, J.M.; Sidney, S.; Reis, J.P. The coronary artery risk development in young adults (CARDIA) study: JACC focus seminar 8/8. *J. Am. Coll. Cardiol.* **2021**, *78*, 260–277. [CrossRef]
74. DeSalvo, K.B.; Bloser, N.; Reynolds, K.; He, J.; Muntner, P. Mortality prediction with a single general self-rated health question: A meta-analysis. *J. Gen. Intern. Med.* **2006**, *21*, 267–275. [CrossRef]
75. Mavaddat, N.; Parker, R.A.; Sanderson, S.; Mant, J.; Kinmonth, A.L. Relationship of self-rated health with fatal and non-fatal outcomes in cardiovascular disease: A systematic review and meta-analysis. *PLoS ONE* **2014**, *9*, e103509. [CrossRef]
76. Lei, L.; Tang, Y.; Zhang, Q.; Xiao, M.; Dai, L.; Lu, J.; Lin, X.; Lu, X.; Luo, W.; Pan, J.; et al. The association between the frequency of annual health checks participation and the control of cardiovascular risk factors. *Front. Cardiovasc. Med.* **2022**, *9*, 860503. [CrossRef] [PubMed]
77. Kondo, T.; Nakano, Y.; Adachi, S.; Murohara, T. Effects of tobacco smoking on cardiovascular disease. *Circ. J.* **2019**, *83*, 1980–1985. [CrossRef] [PubMed]
78. Parmar, M.P.; Kaur, M.; Bhavanam, S.; Mulaka, G.S.R.; Ishfaq, L.; Vempati, R.; Kandepi, H.V.; Er, R.; Sahu, S.; Davalgi, S.; et al. A systematic review of the effects of smoking on the cardiovascular system and general health. *Cureus* **2023**, *15*, e38073. [CrossRef] [PubMed]
79. Rao Kondapally Seshasai, S.; Kaptoge, S.; Thompson, A.; Di Angelantonio, E.; Gao, P.; Sarwar, N.; Whincup, P.H.; Mukamal, K.J.; Gillum, R.F.; Holme, I.; et al. Diabetes mellitus, fasting glucose, and risk of cause-specific death. *N. Engl. J. Med.* **2011**, *364*, 829–841.
80. Dabic, P.; Petrovic, J.; Vucurevic, B.; Bucic, A.; Bajcetic, D.; Ilijevski, N.; Sevkovic, M. Caught between heart and limbs: Navigating the treatment of patients with CAD and PAD in an overwhelmed healthcare system. *Angiology* **2025**, *76*, 193–199. [CrossRef]
81. Meyer, P.W.; Anderson, R.; Ker, J.A.; Ally, M.T. Rheumatoid arthritis and risk of cardiovascular disease. *Cardiovasc. J. Afr.* **2018**, *29*, 317–321. [CrossRef]
82. Aviña-Zubieta, J.A.; Choi, H.K.; Sadatsafavi, M.; Etminan, M.; Esdaile, J.M.; Lacaille, D. Risk of cardiovascular mortality in patients with rheumatoid arthritis: A meta-analysis of observational studies. *Arthritis Care Res.* **2008**, *59*, 1690–1697. [CrossRef]
83. Lichtman, J.H.; Froelicher, E.S.; Blumenthal, J.A.; Carney, R.M.; Doering, L.V.; Frasure-Smith, N.; Freedland, K.E.; Jaffe, A.S.; Leifheit-Limson, E.C.; Sheps, D.S.; et al. Depression as a risk factor for poor prognosis among patients with acute coronary syndrome: Systematic review and recommendations: A scientific statement from the American Heart Association. *Circulation* **2014**, *129*, 1350–1369. [CrossRef]
84. Armenian, S.H.; Xu, L.; Ky, B.; Sun, C.; Farol, L.T.; Pal, S.K.; Douglas, P.S.; Bhatia, S.; Chao, C. Cardiovascular disease among survivors of adult-onset cancer: A community-based retrospective cohort study. *J. Clin. Oncol.* **2016**, *34*, 1122–1130. [CrossRef]
85. Suman, S.; Pravalika, J.; Manjula, P.; Farooq, U. Gender and CVD-does it really matters? *Curr. Probl. Cardiol.* **2023**, *48*, 101604. [CrossRef]
86. Maas, A.H.; Appelman, Y.E. Gender differences in coronary heart disease. *Neth. Heart J.* **2010**, *18*, 598–603. [CrossRef]
87. Powell-Wiley, T.M.; Poirier, P.; Burke, L.E.; Després, J.P.; Gordon-Larsen, P.; Lavie, C.J.; Lear, S.A.; Ndumele, C.E.; Neeland, I.J.; Sanders, P.; et al. Obesity and cardiovascular disease: A scientific statement from the American Heart Association. *Circulation* **2021**, *143*, e984–e1010. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.